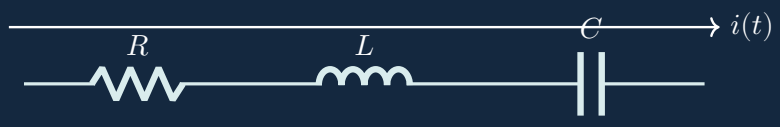
EXAM PREPARATION SERIES / 2026

SSC JE ELECTRICAL MASTER NOTES

BASIC ELECTRICAL & CIRCUIT THEORY



Full derivations | Accurate circuit diagrams | Stepwise solved models

PYQs examined from 2023, 2024 and 2025 | Answers verified independently

Specifically for the SSC JE Electrical 2026 syllabus
Fundamentals, DC analysis, circuit theorems, magnetic circuits, AC, resonance, complex power and balanced three-phase networks. Extra switching and bridge examples are flagged as supporting material.

11 October 2026 Independent study guide (not an official SSC publication)

Read this first

This edition replaces a brief formula-oriented booklet with a *method-first* reference. Every major formula is either derived or connected to its physical meaning; the circuit symbols show exactly which nodes and elements are involved. Numerical problems are grouped by the method SSC JE tests, not merely by formula.

Key result

2026 official scope. The SSC notice, paragraph 14.3.2 (printed page 26) explicitly names circuit elements and units; KCL/KVL, node and mesh analysis; Thevenin, Norton, superposition, maximum power transfer; magnetic flux, reluctance, MMF, self/mutual inductance; RMS/average values, AC series/parallel circuits, resonance, balanced three-phase, star–delta, complex power and power factor.

`https://ssc.gov.in/api/attachment/uploads/masterData/NoticeBoards/Notice_of_adv_je_2026.pdf`

How to study (1) Learn the law, (2) trace the circuit, (3) solve an example without looking, (4) compare with the worked solution, (5) do the end-of-book drills, (6) repeat PYQs after 48 hours.

SSC JE trap / important distinction

PYQ labeling policy: Exam-year/shift/question identifiers in the PYQ chapter refer to publicly available reproductions of the actual past papers. The wording in this book is *paraphrased*; calculations are reworked independently. Where a circulated reproduction is questionable or a figure is unreadable, that item is omitted. Do not confuse the separate original practice exercises with PYQs.

Legend	Meaning
Worked example	Full numerical setup and solution, including common variants
Key result	The fastest reliable equation or invariant for revision
Exam trap	A common incorrect assumption, missing unit conversion, or sign mistake
PYQ / year / shift	Paraphrased, source-checked prior-paper numerical or concept item
Practice drill	Fresh original question, not represented as an actual SSC paper

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Quick notation and units

Symbol	Quantity	Unit
q, i, I	Charge; instantaneous current; current magnitude	C, A
v, V, E	Voltage; voltage magnitude; source EMF	V
R, ρ, σ, G	Resistance; resistivity; conductivity; conductance	Ω, Ω m, S/m, S
C, L, M	Capacitance; self-inductance; mutual inductance	F, H, H
Φ, B, H	Magnetic flux; flux density; field intensity	Wb, T, A/m
$\mathcal{R}, \mu, N$	Reluctance; permeability; number of turns	A-turns/Wb, H/m, –
f, ω, T	Frequency; angular frequency; period	Hz, rad/s, s
Z, Y, X_L, X_C	Impedance; admittance; inductive/capacitive reactance	Ω, S, Ω
P, Q, S	Real power; reactive power; apparent power	W, var, VA

Key result

$$1\,\text{mA} = 10^{-3}\,\text{A}, \quad 1\,\mu\text{F} = 10^{-6}\,\text{F}, \quad 1\,\text{mH} = 10^{-3}\,\text{H},$$
$$1\,\text{kWh} = 3.6 \times 10^{6}\,\text{J}, \quad 1\,\text{cm}^2 = 10^{-4}\,\text{m}^2, \quad \omega = 2\pi f.$$

UNIT 1

Electrical quantities, laws, power and energy

1.1 The physical picture

Electric charge q is measured in coulomb (C). Electric current is the rate of charge flow, not the amount of charge. Conventional current direction is opposite to electron flow in metallic conductors.

$$i(t) = \frac{dq}{dt}, \qquad q(t_2) - q(t_1) = \int_{t_1}^{t_2} i(t)\,dt.$$

Voltage measures energy transferred per unit charge: $v = dw/dq$. A current of one ampere flows if one coulomb crosses a section per second. One volt means one joule per coulomb.

$$p(t) = v(t)i(t) = \frac{dw}{dt}, \qquad w = \int p(t)\,dt.$$

1.1.1 Passive sign convention

When current enters the terminal marked $+$, $p = vi > 0$ denotes **power absorbed**. When it leaves the $+$ terminal, the element supplies vi to the rest of the circuit. A negative computed current or voltage merely means the actual direction/polarity is opposite the assigned reference.



$p = vi > 0$ for an absorbing resistor

1.2 Ohm's law and limits

For an *ohmic* resistor at fixed temperature, $v = iR$. The constitutive equations of an ideal capacitor and inductor are $i = C\,dv/dt$ and $v = L\,di/dt$. Ohm's law is not a universal law for diodes, lamps with changing temperature, or arbitrary nonlinear devices.

Key result

$$P = VI = I^2R = \frac{V^2}{R}, \qquad W = Pt, \qquad \eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\%.$$

For the I^2R and V^2/R forms, V is the voltage across *that resistor*; in AC the magnitudes are RMS values.

Worked example — **Charge and variable current**

An increasing current is $i(t) = 2 + 3t$ A over $0 \le t \le 4$ s. Find total transferred charge.

$$q = \int_0^4 (2 + 3t)\,dt = \left[2t + \tfrac{3}{2}t^2\right]_0^4 = 8 + 24 = 32\,\text{C}.$$

Answer: $\boxed{q = 32\,\text{C}}$.

Worked example **Energy and bill units**

A $1.5\,\text{kW}$ heater operates for 3 hours. Find energy in kWh and MJ.

$$W = 1.5 \times 3 = 4.5\,\text{kWh}; \quad W = 4.5(3.6) = 16.2\,\text{MJ}.$$

Answer: $\boxed{W = 4.5\,\text{kWh} = 16.2\,\text{MJ}}$.

1.3 Element types and network vocabulary

Active source: delivers energy under some conditions; **passive element:** stores/dissipates energy; **linear:** obeys superposition; **bilateral:** same behavior on reversing direction (ideal R, L, C); **unilateral:** direction-dependent (ideal diode). A **node** is a common connection, a **branch** is a path between nodes, a **loop** is any closed route and a **mesh** is a loop containing no other loop in a planar circuit.

1.4 Power balance

Conservation of energy in a lumped circuit implies *sum of absorbed powers = sum of delivered powers*. For any chosen consistent sign convention, $\sum_k p_k(t) = 0$. It is a powerful answer check after solving node voltages and branch currents.

SSC JE trap / important distinction

Do not apply $P = VI$ to sinusoidal *real* power without the factor $\cos\phi$. VI is *apparent* power unless the voltage and current are in phase.

UNIT 2

Resistance, resistivity and DC networks

2.1 Resistance from material and geometry

For a uniform conductor at a fixed temperature,

$$R = \rho\frac{\ell}{A}, \qquad \sigma = \frac{1}{\rho}, \qquad G = \frac{1}{R}.$$

For a wire of diameter d, $A = \pi d^2/4$. Increasing length increases R; increasing cross-sectional area lowers R. Resistivity depends on material and temperature, not conductor length.

2.1.1 Stretching / cutting / splitting models

A wire stretched without loss of material has constant volume $A\ell$. If the length becomes $k\ell$, then $A' = A/k$, giving

$$\boxed{R' = k^2 R} \quad \text{(constant volume and resistivity).}$$

If it is instead cut into n equal pieces, each has resistance R/n. Connecting all n pieces in parallel gives $R_{\text{eq}} = R/n^2$.

Worked example — **Length doubling**

A wire has resistance $5\,\Omega$ and is uniformly stretched to twice its initial length. Assume constant volume and unchanged ρ.

$$R' = 2^2 R = 4(5) = 20\,\Omega.$$

Answer: $\boxed{R' = 20\,\Omega}$.

2.2 Temperature dependence

For a moderate temperature interval, $R_2 = R_1[1 + \alpha_{T_1}(T_2 - T_1)]$. Temperature coefficients referred to different temperatures are not numerically identical:

$$\alpha_{T_2} = \frac{\alpha_{T_1}}{1 + \alpha_{T_1}(T_2 - T_1)}.$$

Typical metals have positive α; carbon and many semiconductors can exhibit a negative temperature coefficient over relevant ranges. The linear law is an approximation, not valid arbitrarily far from the reference temperature.

Worked example — **Resistance at a new temperature**

A coil is $20\,\Omega$ at 20°C, $\alpha_{20} = 0.004/^\circ$C. Find resistance at 70°C.

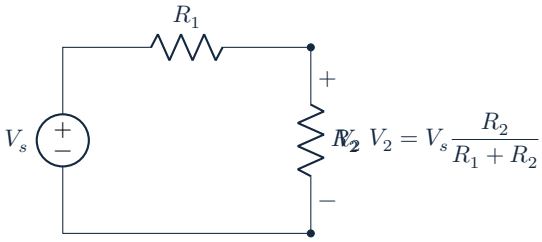
$$R_{70} = 20[1 + 0.004(70 - 20)] = 20(1.2) = 24\,\Omega.$$

Answer: $\boxed{R_{70} = 24\,\Omega}$.

2.3 Series, parallel, voltage division, current division

In series the *same current* flows and the resistances add. In parallel the *same voltage* appears and conductances add.

$$R_{\text{series}} = \sum R_k, \qquad \frac{1}{R_{\text{parallel}}} = \sum \frac{1}{R_k}, \qquad R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}.$$



Voltage divider: the output must be measured across the chosen resistor.

$$V_k = V_s \frac{R_k}{\sum R}, \qquad I_1 = I \frac{R_2}{R_1 + R_2}, \qquad I_2 = I \frac{R_1}{R_1 + R_2}.$$

Worked example **Two-branch current divider**

A total current of 10 A enters parallel branches of $4\,\Omega$ and $6\,\Omega$. Find current in $6\,\Omega$.

$$I_6 = 10 \frac{4}{4+6} = 4\,\text{A}, \qquad I_4 = 6\,\text{A}.$$

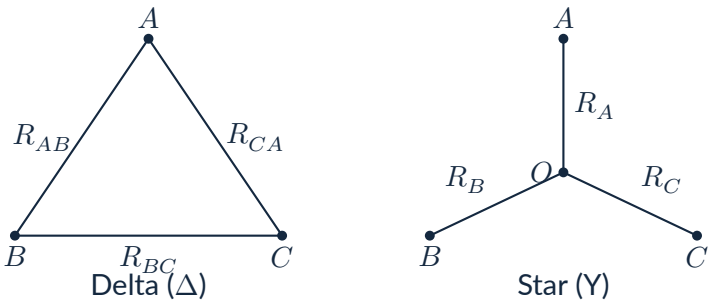
Answer: $\boxed{I_6 = 4\,\text{A}}$.

SSC JE trap / important distinction

Current divides *inversely* with resistance. The branch carrying smaller current has larger resistance. Voltage divides *directly* with resistance in a series circuit.

2.4 Star–delta conversion of resistances

Define delta sides R_{AB}, R_{BC}, R_{CA} and star arms R_A, R_B, R_C to the common center O.



Side and arm subscripts must refer to the same terminals.

Key result

Write $\Sigma_\Delta = R_{AB} + R_{BC} + R_{CA}$. Then

$$\boxed{R_A = \frac{R_{AB}R_{CA}}{\Sigma_\Delta}}, \quad R_B = \frac{R_{AB}R_{BC}}{\Sigma_\Delta}, \quad R_C = \frac{R_{BC}R_{CA}}{\Sigma_\Delta}.$$

Reverse transform:

$$R_{AB} = R_A + R_B + \frac{R_A R_B}{R_C} \quad \text{(cyclically for other sides).}$$

Balanced case: $R_\Delta = 3R_Y$ and $R_Y = R_\Delta/3$.

Worked example **Unequal delta**

A delta has $R_{AB} = 6$, $R_{BC} = 9$, $R_{CA} = 3\ \Omega$. Find the equivalent star.

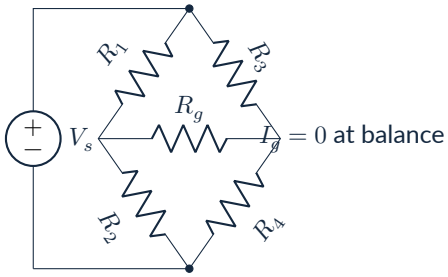
$$\Sigma_\Delta = 18\,\Omega; \quad R_A = \frac{6(3)}{18} = 1,\ R_B = \frac{6(9)}{18} = 3,\ R_C = \frac{9(3)}{18} = 1.5 \quad \Omega.$$

Answer: $\boxed{(R_A, R_B, R_C) = (1, 3, 1.5)\,\Omega}$.

2.5 Wheatstone bridge and circuit symmetry

A bridge has two voltage-dividing arms. With supply between top/bottom and a detector linking the midpoints, the detector current is zero when the midpoint potentials match:

$$\boxed{\frac{R_1}{R_2} = \frac{R_3}{R_4}}.$$



Bridge branch can be removed only when it really carries zero current.

Worked example **Balanced bridge**

Given $R_1 = 2$, $R_2 = 4$, $R_3 = 3\ \Omega$, determine R_4 for zero bridge current.

$$\frac{2}{4} = \frac{3}{R_4} \Longrightarrow R_4 = 6\,\Omega.$$

Answer: $\boxed{R_4 = 6\,\Omega}$.

2.6 Resistor colour code (supporting fundamentals)

For a four-band resistor: first two bands are significant digits, third is multiplier, fourth tolerance. Digits: black 0, brown 1, red 2, orange 3, yellow 4, green 5, blue 6, violet 7, grey 8, white 9. Gold tolerance $\pm 5\%$, silver $\pm 10\%$; no fourth band usually $\pm 20\%$.

Worked example **Colour-code reading**

Red–red–orange–gold $\Rightarrow 22 \times 10^3\ \Omega$ with $\pm 5\%$ tolerance.
Answer: $\boxed{R = 22\,\text{k}\Omega\ \pm 5\%}$.

SSC JE trap / important distinction

Never convert a Δ network into three resistors of $R/3$ unless all three delta sides are equal. The relation is *not* valid for an unbalanced delta.

UNIT 3

Capacitors and electrostatic circuits

3.1 Meaning and construction

Capacitance is stored electric charge per applied potential difference:

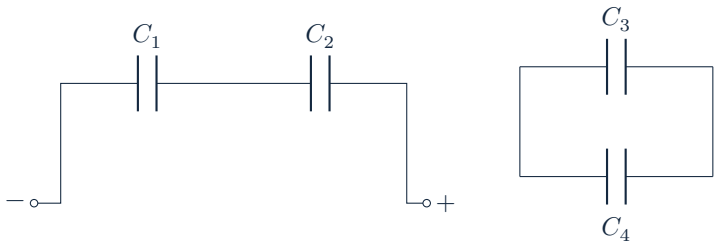
$$q = Cv, \qquad C = \frac{q}{v}, \qquad i(t) = C\frac{dv(t)}{dt}.$$

For two large parallel plates separated by a uniform dielectric (d small relative to lateral plate size),

$$\boxed{C = \frac{\varepsilon A}{d}}, \qquad \varepsilon = \varepsilon_0 \varepsilon_r, \quad \varepsilon_0 \approx 8.854 \times 10^{-12}\,\mathrm{F/m}.$$

Increase area or permittivity $\Rightarrow$ increase C; increase separation $\Rightarrow$ decrease C. Breakdown voltage depends on dielectric strength and plate spacing. Leakage and ESR are practical departures from the ideal model.

3.2 Series and parallel capacitor networks



Series Parallel

Series capacitors carry the same magnitude of charge; inverse capacitances add. Parallel capacitors have the same voltage; capacitances add.

$$\boxed{\frac{1}{C_{\text{series}}} = \sum_k \frac{1}{C_k}}, \qquad \boxed{C_{\text{parallel}} = \sum_k C_k}.$$

For series C_1, C_2: $V_1/V_2 = C_2/C_1$. For parallel capacitors: $Q_1/Q_2 = C_1/C_2$.

Worked example — **Charge and voltage in a series bank**

$C_1 = 3\,\mu\mathrm{F}$ and $C_2 = 6\,\mu\mathrm{F}$ are in series across 18 V.

$$C_{eq} = \frac{3 \cdot 6}{3+6} = 2\,\mu\mathrm{F}, \qquad Q = C_{eq}V = 36\,\mu\mathrm{C}.$$

$$V_1 = Q/C_1 = 12\,\mathrm{V}, \quad V_2 = Q/C_2 = 6\,\mathrm{V}.$$

Answer: $\boxed{C_{eq} = 2\,\mu\mathrm{F}, \quad (V_1, V_2) = (12, 6)\,\mathrm{V}}$.

3.3 Energy storage and electric field

Incremental energy supplied $dw = v\,dq = (q/C)dq$. Thus

$$\boxed{W_C = \int_0^Q \frac{q}{C}\,dq = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV}.$$

For ideal parallel plates $E = V/d$ (away from edge fringing) and electric energy density is $u_e = \frac{1}{2}\varepsilon E^2$ J/m^3.

Worked example **Stored energy**

A $20\,\mu$F capacitor is charged to 100 V.

$$W = \tfrac{1}{2}(20 \times 10^{-6})(100)^2 = 0.10\,\text{J}.$$

Answer: $\boxed{W = 0.10\,\text{J}}$.

3.4 Dielectric inserted: two boundary conditions

Suppose a dielectric fills the gap and $C' = kC$ where $k = \varepsilon_r$.

Quantity	Connected to ideal voltage source	Disconnected and isolated
Held constant	V	Q
New charge	$Q' = kQ$	$Q' = Q$
New voltage	$V' = V$	$V' = V/k$
Stored energy	$W' = kW$	$W' = W/k$

SSC JE trap / important distinction

The energy trends reverse depending on the connection. A source can do work and supply extra charge in the connected case; an isolated capacitor cannot acquire extra net free charge.

3.5 Partial dielectrics and geometry changes

For dielectric slabs laid *side by side* across fractions of plate area (same gap), use **parallel capacitors**: $C = \sum_i \varepsilon_i A_i/d$. For stacked dielectric layers in the *field direction*, use **series capacitors**:

$$\boxed{C = \frac{A}{\sum_i d_i/\varepsilon_i}}.$$

Worked example **Two dielectric layers**

Two layers, each 1 mm thick with relative permittivities 2 and 4, fill a 100 cm^2 capacitor.

$$C = \frac{\varepsilon_0 A}{d_1/2 + d_2/4} = \frac{8.854 \times 10^{-12}(10^{-2})}{(10^{-3}/2) + (10^{-3}/4)} \approx 118\,\text{pF}.$$

Answer: $\boxed{C \approx 118\,\text{pF}}$.

3.6 Capacitor under DC and AC

An ideal capacitor prohibits an instantaneous voltage jump unless subjected to an impulse of current: $v_C(0^+) = v_C(0^-)$. With steady DC, $dv_C/dt = 0$, hence $i_C = 0$; the ideal capacitor behaves as an *open circuit*. With sinusoidal voltage, $Z_C = 1/(\mathrm{j}\omega C) = -\mathrm{j}/(\omega C)$; capacitor current leads voltage by 90°.

Worked example **A capacitor with a ramp voltage**

For $C = 5\,\mu$F, $v(t) = 40t$ V while the ramp is active.

$$i = C\frac{dv}{dt} = 5 \times 10^{-6} \cdot 40 = 200\,\mu\text{A}.$$

Answer: $\boxed{i = 200\,\mu\text{A}}$.

UNIT 4

Inductors, coupled coils and induced EMF

4.1 Inductor definition and energy

Flux linkage $\lambda = N\Phi$ is the sum of flux linkages of the turns. For a *linear* inductor,

$$\lambda = Li, \qquad v = \frac{d\lambda}{dt} = L\frac{di}{dt}, \qquad W_L = \frac{1}{2}Li^2.$$

In a solenoid, increasing turns N, permeability μ, or cross-sectional area A increases inductance; increasing magnetic path length ℓ decreases it, assuming linear material and simple geometry:

$$\boxed{L \simeq \frac{\mu N^2 A}{\ell}}.$$

An ideal inductor resists sudden *change* in current, not current itself. For ordinary finite applied voltage, $i_L(0^+) = i_L(0^-)$. In steady DC, $di/dt = 0$ and ideal $v_L = 0$; it acts as a short circuit.

Worked example — **Energy in a solenoid**

Find magnetic energy for $L = 4$ H and $I = 15$ A.

$$W_L = \tfrac{1}{2}(4)(15)^2 = 450\,\text{J}.$$

Answer: $\boxed{W_L = 450\,\text{J}}$.

4.2 Inductors in series and parallel

Uncoupled series inductors add: $L_{eq} = \sum L_i$. Uncoupled parallel inductors follow $1/L_{eq} = \sum 1/L_i$ provided they share the same applied voltage and their initial conditions are treated consistently. Coupling changes the answer.

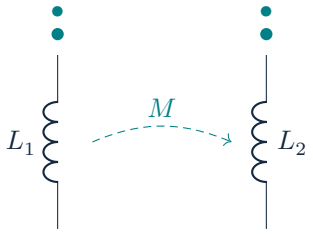
4.3 Mutual inductance and coupling coefficient

If current in coil 1 links coil 2, an induced voltage appears. With consistent reference dots:

$$v_1 = L_1\frac{di_1}{dt} \pm M\frac{di_2}{dt}, \qquad v_2 = L_2\frac{di_2}{dt} \pm M\frac{di_1}{dt}.$$

Coupling coefficient k satisfies $0 \le k \le 1$ for passive coils:

$$\boxed{M = k\sqrt{L_1 L_2}}, \qquad M_{\text{max}} = \sqrt{L_1 L_2}.$$



Dot convention: mutual induced polarity

Dot-rule interpretation: when both reference currents enter dotted ends, the mutual term is aiding. If one enters a dotted end and the other leaves a dotted end, it is opposing for the chosen references.

4.4 Series aiding and opposing

$$\boxed{L_{\text{aiding}} = L_1 + L_2 + 2M}, \qquad \boxed{L_{\text{opposing}} = L_1 + L_2 - 2M}.$$

Subtracting eliminates the unknown self-inductances:

$$\boxed{M = \frac{L_{\text{aiding}} - L_{\text{opposing}}}{4}}.$$

Adding gives $L_1 + L_2 = (L_{\text{aiding}} + L_{\text{opposing}})/2$.

Worked example — **Coupled-coil difference shortcut**

Two measured equivalent series inductances are 150 mH (aiding) and 80 mH (opposing).

$$M = \frac{150 - 80}{4} = 17.5\,\text{mH}.$$

Answer: $\boxed{M = 17.5\,\text{mH}}$.

Worked example — **Maximum coupling**

$L_1 = 20$ H, $L_2 = 320$ H; maximum physically possible M occurs at $k = 1$.

$$M_{max} = \sqrt{20 \cdot 320} = \sqrt{6400} = 80\,\text{H}.$$

Answer: $\boxed{M_{max} = 80\,\text{H}}$.

4.5 Faraday's and Lenz's laws

A flux change induces an EMF opposing the *change* in linked flux (Lenz):

$$\boxed{e = -N\frac{d\Phi}{dt}}.$$

Magnitude form for a uniform flux change: $|E_{avg}| = N|\Delta\Phi|/\Delta t$. If the flux linkage is nonlinear, the incremental inductance is $d\lambda/di$, and the simple energy $\frac{1}{2}Li^2$ requires constant L.

Worked example **Changing flux**

A 200-turn coil sees a flux drop from 0.020 Wb to 0.005 Wb uniformly in 0.10 s.

$$|E| = 200\frac{0.020 - 0.005}{0.10} = 30\,\text{V}.$$

Answer: $\boxed{|E| = 30\,\text{V}}$.

4.6 Magnetic energy with coupled coils

For linear coupled coils in consistently defined passive directions,

$$W = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 \pm M i_1 i_2.$$

The sign depends on dot orientation / selected current reference, not solely on whether both currents are positive numbers. Physical passivity requires $M^2 \le L_1 L_2$.

SSC JE trap / important distinction

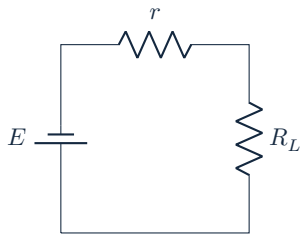
If an SSC question gives two *equivalent inductances* for aiding/opposing, their difference is $4M$, not $2M$. Also k can never be more than 1 in a passive two-coil model.

UNIT 5

Batteries, practical sources and source conversion

5.1 Ideal versus real sources

An ideal voltage source maintains prescribed voltage regardless of load current. A practical battery is modeled by an ideal EMF E with internal series resistance r. An ideal current source maintains prescribed current regardless of terminal voltage; a practical current source includes shunt resistance.



Discharging battery: $V_t = E - Ir$

$$I = \frac{E}{r + R_L}, \qquad V_t = IR_L = E - Ir.$$

During charging, with reference charging current driven into the positive battery terminal, $V_t = E + Ir$ in the same simple model.

Worked example — **Battery terminal voltage**

A 12 V source has $r = 0.1\,\Omega$ and powers $R_L = 1.9\,\Omega$.

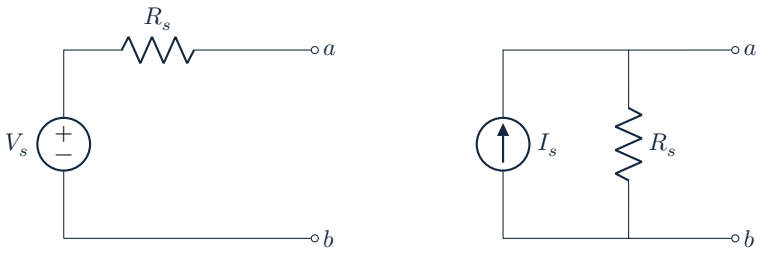
$$I = \frac{12}{2} = 6\,\text{A}, \quad V_t = 12 - 6(0.1) = 11.4\,\text{V}.$$

Answer: $\boxed{V_t = 11.4\,\text{V}}$.

5.2 Batteries in series and parallel

Series-aiding equal batteries: $E_{eq} = nE$, $r_{eq} = nr$; parallel identical batteries: $E_{eq} = E$, $r_{eq} = r/n$. For unlike batteries connected in parallel, circulation currents are possible even without an external load; don't use the identical-battery shortcut blindly. Battery capacity is commonly specified in Ah (charge): $Q \approx It$ at the specified discharge conditions. Energy in Wh is approximately average terminal voltage times Ah; voltage can vary during discharge.

5.3 Source transformation



Thevenin form Norton form

$$\boxed{I_s = \frac{V_s}{R_s}}, \qquad \boxed{V_s = I_s R_s}; \quad R_s \text{ unchanged.}$$

The equivalence is between the *terminal* v–i relationships. Do not transform an isolated ideal voltage source directly into a finite ideal current source when no series resistance is present.

Worked example **Transform a practical source**

A 30 V source in series with $5\,\Omega$ is converted to Norton form.

$$I_N = 30/5 = 6\,\text{A}, \qquad R_N = 5\,\Omega \text{ (in parallel).}$$

Answer: $\boxed{I_N = 6\,\text{A},\ R_N = 5\,\Omega}$.

5.4 Internal loss, load efficiency and power matching

For a source EMF E and resistance r feeding R_L,

$$P_L = I^2 R_L = \frac{E^2 R_L}{(r + R_L)^2}, \quad \eta = \frac{P_L}{EI} = \frac{R_L}{r + R_L}.$$

At $R_L = r$, $\eta = 50\%$ in this idealized model. *Maximum output power* does not mean maximum efficiency.

SSC JE trap / important distinction

An ideal battery's EMF (open-circuit voltage) and terminal voltage under load are not the same if its internal resistance is nonzero. A battery can have a nominal voltage different from its operating terminal voltage.

UNIT 6

KCL, KVL, node analysis and mesh analysis

6.1 Kirchhoff's current law (KCL)

At any node, algebraic current sum is zero ($\sum i = 0$), a consequence of charge conservation in lumped circuits. Choose currents entering as positive, or leaving as positive, consistently.

Worked example — **One-node current balance**

At a junction, 5 A and 3 A enter; 2 A and unknown I leave. KCL: $5 + 3 = 2 + I$.
Answer: $\boxed{I = 6\,\text{A}}$.

6.2 Kirchhoff's voltage law (KVL)

Around a closed loop, algebraic voltage sum equals zero, provided the loop voltage definitions account for any time-varying linked flux. For standard lumped circuit diagrams, this is equivalent to conservation of energy.

$$\sum_{\text{loop}} v_k = 0.$$

If you traverse through a resistor in the current direction, write a voltage drop $-IR$ when adding rises; across a source from negative to positive terminal, add $+E$.

Worked example — **Single-loop KVL**

A 24 V ideal source drives series $4\,\Omega$ and $8\,\Omega$ resistors.

$$24 - 4I - 8I = 0 \;\Rightarrow I = 2\,\text{A}.$$

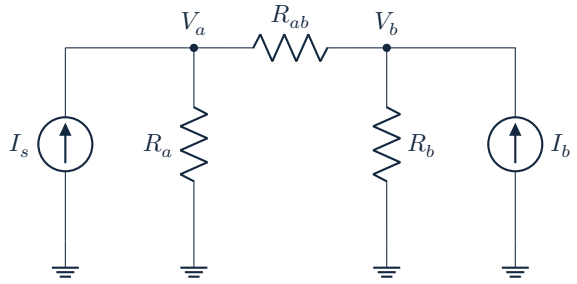
The resistor drops are 8 V and 16 V; $8 + 16 = 24$ V verifies KVL.
Answer: $\boxed{I = 2\,\text{A}}$.

6.3 Nodal analysis: the universal workflow

Choose one node as ground (0 V). Assign voltages $V_1, V_2, \ldots$ to the remaining nodes. For each unknown node, sum resistor currents *leaving*:

$$\sum_k \frac{V_i - V_k}{R_{ik}} = I_{\text{injected},i}.$$

Equivalent conductance form $G\,\mathbf{V} = \mathbf{I}$ follows; diagonal of G sums conductances connected to that node, off-diagonal terms are negative mutual conductances.



Reference node is at ground; label each unknown potential before writing KCL.

For the diagram, if both arrows inject current upward: $\frac{V_a}{R_a} + \frac{V_a - V_b}{R_{ab}} = I_s$, and $\frac{V_b}{R_b} + \frac{V_b - V_a}{R_{ab}} = I_b$.

Solve systematically

1. Identify the reference and label all non-reference nodes.
2. Write each branch current as $(V_{\text{here}} - V_{\text{other}})/R$.
3. Sum leaving currents and equate to injected source current.
4. Solve simultaneous equations; a negative branch current means reverse actual flow.
5. Recheck KCL numerically at each node.

Worked example **Two-node network**

Node a connects to ground by $2\,\Omega$, node b to ground by $4\,\Omega$, and a to b by $2\,\Omega$. A 6 A source injects at a; none at b.

$$\frac{V_a}{2} + \frac{V_a - V_b}{2} = 6 \quad \Rightarrow 2V_a - V_b = 12,$$
$$\frac{V_b}{4} + \frac{V_b - V_a}{2} = 0 \quad \Rightarrow -2V_a + 3V_b = 0.$$

From the second, $V_a = 1.5V_b$. Substitute: $3V_b - V_b = 12$, so $V_b = 6$ V and $V_a = 9$ V. Current $a \to b = (9-6)/2 = 1.5$ A.

Answer: $\boxed{V_a = 9\,\text{V},\ V_b = 6\,\text{V},\ I_{a\to b} = 1.5\,\text{A}}$.

6.4 Supernodes and dependent sources

If an ideal voltage source sits between two non-reference nodes a, b, form a **supernode**. Apply KCL to the outside of the combined pair, then add its voltage constraint $V_a - V_b = E$ according to polarity. A voltage source between one node and ground simply fixes that node's voltage.

Worked example **Supernode**

Two nodes a, b have a 10 V source with $V_a - V_b = 10$ V. Node a connects to ground by $5\,\Omega$ and node b by $10\,\Omega$. A 3 A source injects into the supernode.

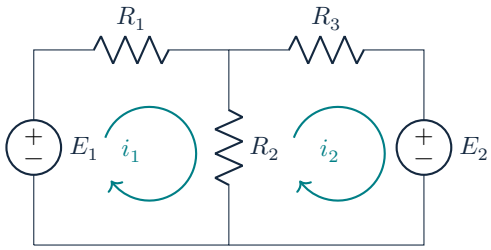
$$\frac{V_a}{5} + \frac{V_b}{10} = 3, \quad V_a - V_b = 10.$$

Multiply KCL by 10: $2V_a + V_b = 30$. Solve with $V_b = V_a - 10$: $3V_a = 40$, giving $V_a = 40/3$ V, $V_b = 10/3$ V.

Answer: $\boxed{V_a = 13.33\,\text{V},\ V_b = 3.33\,\text{V}}$.

6.5 Mesh analysis (planar networks)

A mesh is an elementary planar loop. Assign all mesh currents, usually clockwise. If a resistor belongs to two adjacent meshes i_1, i_2, its voltage drop in mesh 1 is $R(i_1 - i_2)$. Write KVL for each mesh, respecting source polarities.



The shared resistor carries $i_1 - i_2$ in the direction of mesh 1.

Worked example **Two-loop equations**

A 10 V driving source acts in mesh 1, a 4 V driving source in mesh 2. Each mesh has one independent $2\,\Omega$ resistor and a common $2\,\Omega$ resistor. Source orientations support clockwise currents.

$$2i_1 + 2(i_1 - i_2) = 10 \quad \Rightarrow 4i_1 - 2i_2 = 10,$$
$$2i_2 + 2(i_2 - i_1) = 4 \quad \Rightarrow -2i_1 + 4i_2 = 4.$$

Solve first for $2i_1 - i_2 = 5$, so $i_2 = 2i_1 - 5$. Substitute: $-2i_1 + 8i_1 - 20 = 4 \Rightarrow i_1 = 4$ A, $i_2 = 3$ A. Shared-branch current $= i_1 - i_2 = 1$ A.
Answer: $\boxed{i_1 = 4\,\text{A},\ i_2 = 3\,\text{A}}$.

6.6 Supermesh

If a current source lies between two meshes, remove its branch from the perimeter when writing the *supermesh KVL*; add the source's constraint, for example $i_2 - i_1 = I_s$ if the arrow points in the direction of mesh 2 through that branch. If the source is on an outside edge, it directly sets that mesh current up to sign.

6.7 Dependent sources must stay active

A dependent source is controlled by a variable elsewhere in the circuit: μv_x (VCVS), $g_m v_x$ (VCCS), $r_m i_x$ (CCVS), βi_x (CCCS). In source deactivation for equivalent resistance, *only independent sources* are set to zero. With dependent sources use a test current/voltage at the output terminals and compute $R_{th} = V_{test}/I_{test}$.

SSC JE trap / important distinction

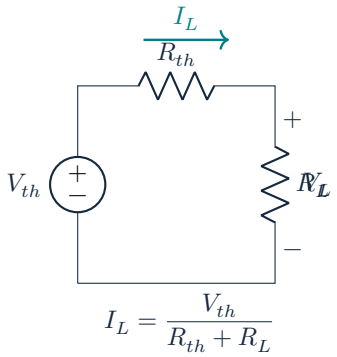
A voltage source between two unknown nodes calls for a *supernode*; a current source between two meshes calls for a *supermesh*. The two ideas are not interchangeable.

UNIT 7

Network theorems and equivalent circuits

7.1 Thevenin's theorem: what it actually says

Any two-terminal *linear* network can be represented externally as an ideal voltage source V_{th} in series with R_{th} (for DC resistive networks) or Z_{th} (AC phasors). The voltage V_{th} is the open-circuit terminal voltage. For ordinary independent-source-only networks, deactivate independent voltage sources (short them) and current sources (open them) to find R_{th}.



Thevenin replacement as seen at the load terminals.

Solve systematically

1. Define terminals a, b across the *removed* load; note voltage polarity.
2. Open the load and calculate V_{ab} using KCL/KVL: this is V_{th}.
3. Determine R_{th} by zeroing *independent* sources only (or apply a test source if dependent sources occur).
4. Reconnect the load to the equivalent source and series resistance.
5. Check whether your result has the expected polarity and units.

Worked example — **Thevenin with voltage divider**

An ideal 20 V source supplies $R_1 = 4\,\Omega$ in series with a branch $R_2 = 6\,\Omega$ to ground. A variable load attaches in parallel with R_2; find equivalent seen by the load.
Step 1. With load open, $V_{th} = 20 \cdot \frac{6}{4+6} = 12$ V.
Step 2. Short the 20 V ideal voltage source: the load terminals see $4 \parallel 6 = 24/10 = 2.4\,\Omega$.
Step 3. For $R_L = 3.6\,\Omega$, $I_L = 12/(2.4 + 3.6) = 2$ A.
Answer: $\boxed{V_{th} = 12\,\text{V},\ R_{th} = 2.4\,\Omega,\ I_L = 2\,\text{A}}$.

7.2 Norton's theorem

Norton form is an ideal current source I_N in parallel with R_N where $I_N = I_{sc}$, the terminal *short-circuit current*; $R_N = R_{th}$ for the same network.

$$\boxed{V_{th} = I_N R_N}, \qquad \boxed{I_N = V_{th}/R_{th}}.$$

With load R_L, current division yields

$$I_L = I_N \frac{R_N}{R_N + R_L}.$$

Worked example **Norton to Thevenin**

The Norton current is 5 A and the parallel resistance is $4\,\Omega$. Therefore $V_{th} = 5 \times 4 = 20$ V, $R_{th} = 4\,\Omega$.
Answer: $\boxed{V_{th} = 20\,\text{V}}$.

7.3 Source-deactivation table

Source	Set to zero means	Equivalent when deactivated
Independent ideal voltage source	$V = 0$ V	Short circuit ($0\,\Omega$)
Independent ideal current source	$I = 0$ A	Open circuit (infinite resistance)
Dependent voltage/current source	Keep active	Use test source, $R_{th} = V_t/I_t$
Real source with internal resistance	Zero only its ideal source	Internal resistor remains

SSC JE trap / important distinction

Never short a current source or open a voltage source to deactivate it; these are the two most common mistakes. For a dependent-source network, short-circuit current and open-circuit voltage can be used to infer resistance only when the relationship is well-defined.

7.4 Superposition theorem

In a *linear* network, voltage or current response with several independent sources equals the algebraic sum of separate responses. Analyze one independent source at a time; replace other *independent* voltage sources by shorts and other *independent* current sources by opens, preserving dependent sources. Responses must be added with signs and reference directions.

Worked example **Two sources, two contributions**

A branch resistance of $5\,\Omega$ sees opposing ideal voltage sources of 20 V and 5 V in series. Find current from superposition.

$$I_{20} = 20/5 = 4\,\text{A}, \quad I_5 = -5/5 = -1\,\text{A}.$$

$$I = 4 - 1 = 3\,\text{A}.$$

Answer: $\boxed{I = 3\,\text{A}}$.

SSC JE trap / important distinction

Do not superpose powers! Power is nonlinear in voltage/current ($P = I^2R$). First add signed voltages/currents; *then* calculate P. Superposition does not directly apply to a nonlinear diode network.

7.5 Maximum power transfer: DC

Replace the source network by V_{th}, R_{th}. Load power is

$$P_L = \frac{V_{th}^2 R_L}{(R_{th} + R_L)^2}.$$

Set the derivative to zero:

$$\frac{dP_L}{dR_L} = V_{th}^2 \frac{R_{th} - R_L}{(R_{th} + R_L)^3} = 0 \quad \Rightarrow \quad \boxed{R_L = R_{th}}.$$

Then $P_{max} = V_{th}^2/(4R_{th})$ for $R_{th} > 0$.

Worked example **Power-maximising load**

A network reduces to $V_{th} = 24$ V, $R_{th} = 6\,\Omega$. Find best resistive load and its power.

$$R_L = 6\,\Omega, \qquad P_{max} = \frac{24^2}{4 \cdot 6} = 24\,\text{W}.$$

Answer: $\boxed{R_L = 6\,\Omega,\ P_{max} = 24\,\text{W}}$.

7.6 AC load matching

For $Z_{th} = R_{th} + \text{j}X_{th}$ with a *freely selectable complex load*, maximum average power occurs at

$$\boxed{Z_L = Z_{th}^* = R_{th} - \text{j}X_{th}} \quad \text{(conjugate match)}.$$

If the load must be purely resistive, $P_L \propto R_L/[(R_{th} + R_L)^2 + X_{th}^2]$, yielding

$$\boxed{R_L = |Z_{th}| = \sqrt{R_{th}^2 + X_{th}^2}}.$$

Do not confuse the two conditions. In typical SSC numerical questions, note whether the load is constrained to a pure resistor.

7.7 Helpful related methods (extension)

Millman's theorem: for parallel branches comprising an ideal voltage source E_k in series with R_k, with consistent EMF orientation,

$$V = \frac{\sum E_k/R_k}{\sum 1/R_k}.$$

Reciprocity: for an appropriate linear bilateral single-independent-source network, the excitation and response locations can be exchanged with the corresponding response relationship. It is not generally valid for arbitrary dependent/nonlinear or non-reciprocal devices.

Worked example **Two parallel source branches**

Parallel practical sources: $E_1 = 12$ V with $R_1 = 3\,\Omega$, and $E_2 = 6$ V with $R_2 = 6\,\Omega$ (same polarity). Their no-load terminal voltage is

$$V = \frac{12/3 + 6/6}{1/3 + 1/6} = \frac{5}{1/2} = 10\,\text{V}.$$

Answer: $\boxed{V = 10\,\text{V}}$.

UNIT 8

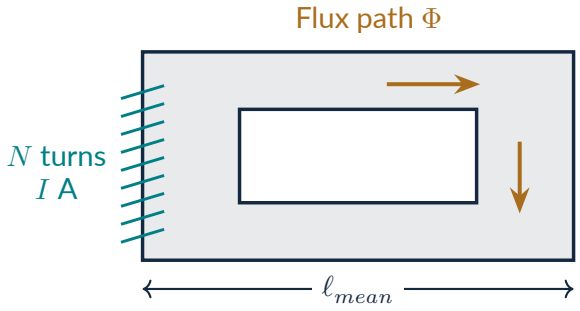
Magnetic circuits and electromagnetic induction

8.1 Flux, flux density and field intensity

Magnetic flux Φ is measured in webers; flux density $B = \Phi/A$ in tesla for uniform normal flux. Field intensity H is measured in ampere-turns per metre (A/m). In a linear, isotropic medium,

$$\boxed{B = \mu H}, \qquad \mu = \mu_0 \mu_r, \qquad \mu_0 = 4\pi \times 10^{-7}\ \text{H/m}.$$

Magnetomotive force is $\mathcal{F} = NI$ (ampere-turns); Ampere's law is $\oint \mathbf{H} \cdot d\ell = NI$. For an approximately uniform magnetic path $H\ell \approx NI$, hence $H \approx NI/\ell$.



Idealised magnetic core: cross-section A, mean path length ℓ.

8.2 Reluctance and magnetic Ohm's law

In a uniform linear section,

$$\boxed{\mathcal{R} = \frac{\ell}{\mu A}}, \qquad \boxed{NI = \Phi \mathcal{R}}, \qquad \boxed{\Phi = \frac{NI}{\mathcal{R}}}.$$

Reluctances in series add for a common flux, whereas reluctance branches in parallel combine through permeances, the magnetic analogy of electrical conductance. For multiple materials carrying the same flux:

$$NI = \Phi \sum_i \frac{\ell_i}{\mu_i A_i}.$$

This is an idealised magnetic-circuit approximation; flux leakage, fringing, saturation, and hysteresis change real systems.

Worked example — **MMF from a known flux**

A core reluctance is 3×10^5 A-turns/Wb and the required flux is 2 mWb.

$$NI = \Phi \mathcal{R} = (0.002)(3 \times 10^5) = 600\ \text{A·turns}.$$

For $N = 300$ turns, $I = 600/300 = 2$ A.
Answer: $\boxed{NI = 600\ \text{A·turns},\ I = 2\ \text{A}}$.

8.3 Air gaps: why MMF rises sharply

Since air has low permeability compared with ferromagnetic core material, its reluctance can dominate despite short gap length. For uniform cross-section and negligible fringing:

$$\mathcal{R}_g = \frac{g}{\mu_0 A}, \qquad NI = H_c \ell_c + H_g g = \frac{B\ell_c}{\mu_c} + \frac{Bg}{\mu_0}.$$

Worked example **Air-gap MMF**

An air gap has $g = 1.0$ mm, area $A = 5\,\text{cm}^2$, and flux $\Phi = 0.5$ mWb. Ignore fringing and core drop.

$$B = \frac{0.5 \times 10^{-3}}{5 \times 10^{-4}} = 1\,\text{T}, \quad NI \approx \frac{Bg}{\mu_0} = \frac{10^{-3}}{4\pi \times 10^{-7}} \approx 796\ \text{A·turns}.$$

Answer: $\boxed{NI \approx 796\ \text{A·turns}}$.

8.4 Self-inductance via reluctance

Since $\lambda = N\Phi = \frac{N^2 I}{\mathcal{R}}$ in a linear circuit,

$$\boxed{L = \frac{\lambda}{I} = \frac{N^2}{\mathcal{R}} = \frac{\mu N^2 A}{\ell}}.$$

Worked example **Inductance of a core**

$N = 400$, $\mu_r = 1500$, $\ell = 0.30$ m, $A = 5\,\text{cm}^2$. Assuming linear uniform flux,

$$L = \frac{4\pi \times 10^{-7}(1500)(400)^2(5 \times 10^{-4})}{0.30} = 0.503\,\text{H} \quad \text{(approx)}.$$

Answer: $\boxed{L \approx 0.503\,\text{H}}$.

8.5 Series/parallel flux splits

In a series magnetic path, a common flux flows through sections and MMF drops add. In parallel branches, MMF between common magnetic nodes is equal (ideal model), while total flux is the sum of branch fluxes. Think $\mathcal{F} \leftrightarrow V$, $\Phi \leftrightarrow I$, $\mathcal{R} \leftrightarrow R$; the analogy is useful but not literal in the presence of leakage and nonlinear materials.

8.6 B–H curve, saturation and losses

A ferromagnetic material generally exhibits nonlinear B–H response. Saturation reduces incremental permeability; higher exciting current is needed for a modest additional flux increase. **Retentivity** is retained B after removal of H, and **coercivity** is reverse field needed to bring B to zero. Area inside a complete B–H hysteresis loop equals energy loss per unit volume per cycle (under the relevant units/-conditions). Eddy-current loss is reduced by laminations and high-resistivity cores.

SSC JE trap / important distinction

Magnetic reluctance is not electric resistance. Ampere-turns are magnetomotive force, not volts; flux in Wb is not current in amperes. When computing $B = \Phi/A$, change cm^2 into m^2 before substituting.

UNIT 9

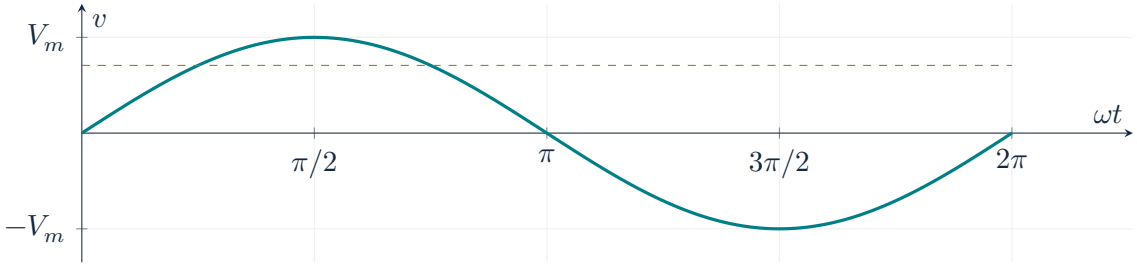
AC waveforms, RMS, average value and phasors

9.1 The sinusoid and its basic parameters

An alternating waveform is often specified by

$$v(t) = V_m \sin(\omega t + \theta), \qquad \omega = 2\pi f, \quad T = 1/f.$$

V_m is peak (amplitude), $V_{pp} = 2V_m$ peak-to-peak, f frequency, T time period, and θ initial phase angle. Electrical angle and time relate through $\theta = \omega t$. A full cycle is 2π rad $= 360^\circ$.



Sine-wave amplitude and RMS level: $V_{rms} \approx 0.707 V_m$.

9.2 Mean and RMS definitions (not memorized slogans)

For any periodic $x(t)$ over a complete period T,

$$x_{avg} = \frac{1}{T}\int_0^T x(t)\,dt, \qquad x_{rms} = \sqrt{\frac{1}{T}\int_0^T [x(t)]^2\,dt}.$$

RMS is defined by *equal heating effect*: $I_{rms}^2 R$ equals the average heating power of a time-varying current in the same resistor. For a pure sine wave,

$$V_{rms} = \frac{V_m}{\sqrt{2}} = 0.7071 V_m, \quad V_{avg,full\ cycle} = 0, \quad V_{avg,half\ cycle} = \frac{2V_m}{\pi}.$$

A rectified sine has nonzero mean: half-wave rectified sine $V_{avg} = V_m/\pi$, $V_{rms} = V_m/2$; full-wave rectified sine $V_{avg} = 2V_m/\pi$, $V_{rms} = V_m/\sqrt{2}$.

Waveform (peak X_m)	Mean over full period	RMS
Pure sine, $X_m \sin \omega t$	0	$X_m/\sqrt{2}$
Half-wave rectified sine	X_m/π	$X_m/2$
Full-wave rectified sine	$2X_m/\pi$	$X_m/\sqrt{2}$
Symmetric square wave $\pm X_m$	0	X_m
Unipolar square wave, 50% duty	$X_m/2$	$X_m/\sqrt{2}$
Symmetric triangular wave $\pm X_m$	0	$X_m/\sqrt{3}$

9.3 Form factor and peak factor

The conventional AC form factor uses the *average of the rectified waveform*:

$$k_f = \frac{X_{rms}}{X_{avg,rectified}}, \qquad k_p = \frac{X_m}{X_{rms}}.$$

For a sine wave: $k_f = \frac{\pi}{2\sqrt{2}} \approx 1.11$, $k_p = \sqrt{2} \approx 1.414$.

Worked example **Peak from an RMS rating**

A sinusoidal supply has 230 V RMS. Find maximum voltage and peak-to-peak voltage.

$$V_m = 230\sqrt{2} \approx 325.3\,\text{V}, \qquad V_{pp} \approx 650.5\,\text{V}.$$

Answer: $\boxed{V_m \approx 325\,\text{V},\ V_{pp} \approx 651\,\text{V}}$.

9.4 Phasors and phase difference

A sine wave $x(t) = \sqrt{2}X\cos(\omega t + \theta)$ corresponds to the RMS phasor $\underline{X} = X\angle\theta = X(\cos\theta + \text{j}\sin\theta)$; phasors apply to *single-frequency steady-state* sinusoids. A $+90^\circ$ rotation is multiplication by j; a -90° rotation is multiplication by $-$j.

$$\underline{V} = \underline{I}\,Z, \qquad Z = R + \text{j}X, \qquad |Z| = \sqrt{R^2 + X^2}, \quad \phi = \tan^{-1}(X/R).$$

If voltage leads current, the load is inductive and $X > 0$; if current leads voltage, it is capacitive and $X < 0$.

Worked example **Phasor interpretation**

If $\underline{V} = 120\angle 30^\circ$ V and $\underline{I} = 5\angle(-10^\circ)$ A, the impedance is $Z = 24\angle 40^\circ\,\Omega$; power factor is $\cos 40^\circ$ lagging because current lags voltage.
Answer: $\boxed{\phi = 40^\circ, \quad \text{pf} \approx 0.766 \text{ lagging}}$.

9.5 Harmonics: RMS values add in squares

For components at distinct integer harmonics of one fundamental frequency, the time-average cross-products vanish over a common period. Thus

$$v(t) = \sum_n V_{m,n}\sin(n\omega t + \theta_n) \quad \Rightarrow \quad \boxed{V_{rms}^2 = \sum_n \frac{V_{m,n}^2}{2}}$$

with any DC component adding V_{DC}^2 separately. This square-sum property applies to different frequencies, *not* to same-frequency phasors (which must first be combined vectorially).

Worked example **Harmonic RMS**

A voltage waveform is $v = 2\sin\omega t + 3\sin 3\omega t + 5\sin 5\omega t$ V.

$$V_{rms} = \sqrt{\frac{2^2 + 3^2 + 5^2}{2}} = \sqrt{19} \approx 4.36\,\text{V}.$$

Answer: $\boxed{V_{rms} = \sqrt{19}\,\text{V} \approx 4.36\,\text{V}}$.

SSC JE trap / important distinction

When a question simply asks "average value of a sinusoidal AC", determine whether it means *whole-cycle average* (0), *half-cycle average* ($2X_m/\pi$), or *rectified average*. The distinction changes the answer.

UNIT 10

Single-phase AC circuits, impedance and admittance

10.1 Three ideal elementary AC elements

Starting with $v = Ri$, $v = L\,di/dt$, and $i = C\,dv/dt$, assume phasors rotating at angular frequency ω. Differentiation multiplies a phasor by $\mathrm{j}\omega$:

$$\boxed{Z_R = R}, \qquad \boxed{Z_L = \mathrm{j}\omega L = \mathrm{j}X_L}, \qquad \boxed{Z_C = \frac{1}{\mathrm{j}\omega C} = -\frac{\mathrm{j}}{\omega C} = -\mathrm{j}X_C}.$$

Here $X_L = 2\pi f L > 0$ and $X_C = 1/(2\pi f C) > 0$ are *positive magnitudes*; the capacitive term enters total reactance with a minus sign.

Property	Pure resistor	Pure inductor	Pure capacitor
Current relative to voltage	In phase	Lags by 90°	Leads by 90°
Impedance angle	0°	$+90^\circ$	-90°
Average real power	I^2R	0 ideal	0 ideal
Reactive power	0	$+I^2X_L$	$-I^2X_C$
If f increases	Unchanged R (ideal)	X_L increases	X_C decreases

Key result

Memory aid: CIVIL. In a **C**apacitor, **I** leads **V**; in an **L** (inductor), **V** leads **I**. This is a mnemonic, not a substitute for phasor convention.

Worked example — **Inductive and capacitive reactance**

At 50 Hz, take $L = 0.1$ H and $C = 100\,\mu$F.

$$X_L = 2\pi(50)(0.1) = 31.42\,\Omega, \qquad X_C = \frac{1}{2\pi(50)(100\times10^{-6})} = 31.83\,\Omega.$$

Answer: $\boxed{X_L \approx 31.42\,\Omega, \quad X_C \approx 31.83\,\Omega}$.

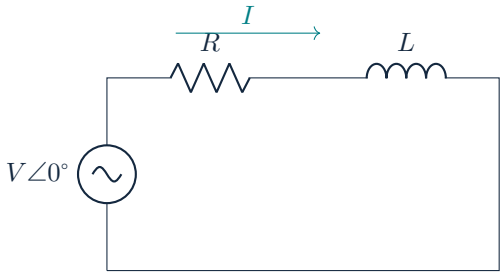
10.2 R–L series circuit

Impedance $Z = R + \mathrm{j}X_L$. For rms voltage V,

$$|Z| = \sqrt{R^2 + X_L^2}, \quad I = \frac{V}{|Z|}, \quad \phi = \tan^{-1}\frac{X_L}{R}, \quad \cos\phi = \frac{R}{|Z|}.$$

Current lags source voltage by ϕ. Resistor and inductor voltages are at right angles in phasor space:

$$V_R = IR, \quad V_L = IX_L, \quad \boxed{V = \sqrt{V_R^2 + V_L^2}}.$$



Series RL: same current through R and L, voltage phasors add.

Worked example **RL: current, power factor, real power**

A 100 V RMS sinusoidal source feeds $R = 6\,\Omega$ and $X_L = 8\,\Omega$ in series.

$$|Z| = \sqrt{6^2 + 8^2} = 10\,\Omega; \quad I = 100/10 = 10\,\text{A}; \quad \text{pf} = 6/10 = 0.6 \text{ lagging}.$$

$$P = I^2R = 10^2(6) = 600\,\text{W}; \quad Q = I^2X_L = 800\,\text{var}.$$

Answer: $\boxed{I = 10\,\text{A},\ P = 600\,\text{W},\ Q = 800\,\text{var}}$.

10.3 R–C series circuit

Impedance $Z = R - \text{j}X_C$ has negative phase angle. Magnitude $|Z| = \sqrt{R^2 + X_C^2}$, phase magnitude $\tan^{-1}(X_C/R)$. Current *leads* voltage. In terms of RMS magnitudes,

$$V = \sqrt{V_R^2 + V_C^2}; \quad \cos|\phi| = R/|Z|.$$

Worked example **RC phase**

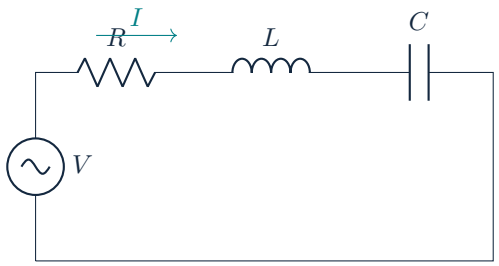
For $R = 8\,\Omega$, $X_C = 6\,\Omega$, $V = 100$ V RMS,

$$Z = 8 - \text{j}6,\ |Z| = 10,\ I = 10\,\text{A},\ \text{pf} = 0.8 \text{ leading}.$$

$$P = 10^2(8) = 800\,\text{W}, \qquad Q = -10^2(6) = -600\,\text{var}.$$

Answer: $\boxed{I = 10\,\text{A},\ \text{pf} = 0.8 \text{ leading},\ Q = -600\,\text{var}}$.

10.4 R–L–C series: the general AC circuit



Series RLC, one branch current, three phasor voltage drops.

$$\boxed{Z = R + \mathrm{j}\left(\omega L - \frac{1}{\omega C}\right)}, \quad \boxed{|Z| = \sqrt{R^2 + (X_L - X_C)^2}}.$$

$$\tan\phi = \frac{X_L - X_C}{R}, \quad \text{pf} = \frac{R}{|Z|}, \quad V = \sqrt{V_R^2 + (V_L - V_C)^2}.$$

The sign of $X_L - X_C$ determines whether current leads or lags.

Worked example **RLC with competing reactances**

For $V = 120$ V, $R = 6\,\Omega$, $X_L = 12\,\Omega$, $X_C = 4\,\Omega$,

$$Z = 6 + \mathrm{j}8\,\Omega, \quad |Z| = 10\,\Omega, \quad I = 12\,\text{A}.$$

$$\cos\phi = 6/10 = 0.6 \text{ lagging}; \quad V_R = 72,\ V_L = 144,\ V_C = 48 \text{ V}.$$

Source $V = \sqrt{72^2 + (144 - 48)^2} = 120$ V, as required.

Answer: $\boxed{I = 12\,\text{A}, \quad \text{pf} = 0.6 \text{ lagging}}$.

10.5 Parallel AC circuits use admittance

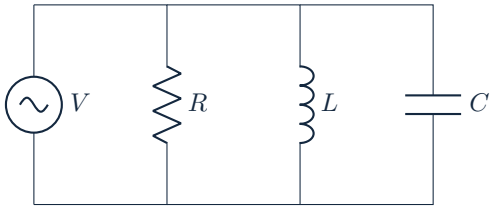
Impedances are easiest in series; **admittances** are easiest in parallel. For one branch $Z = R + \mathrm{j}X$,

$$Y = \frac{1}{Z} = \frac{R - \mathrm{j}X}{R^2 + X^2} = G + \mathrm{j}B.$$

For a parallel R, L, C network,

$$\boxed{Y_{eq} = \frac{1}{R} + \frac{1}{\mathrm{j}\omega L} + \mathrm{j}\omega C} = \frac{1}{R} + \mathrm{j}\left(\omega C - \frac{1}{\omega L}\right).$$

Then $\underline{I} = \underline{V} Y_{eq}$; branch *currents* must add vectorially, not arithmetically.



Parallel RLC: voltage is common, source current is a phasor sum.

Worked example **Parallel RL: current from branch phasors**

A 100 V source is connected in parallel to $R = 20\,\Omega$ and $X_L = 25\,\Omega$.

$$\underline{I}_R = 100/20 = 5\angle 0^\circ\ \text{A}, \quad \underline{I}_L = 100/(\mathrm{j}25) = -\mathrm{j}4\,\text{A}.$$

$$\underline{I} = 5 - \mathrm{j}4, \quad |I| = \sqrt{25 + 16} = \sqrt{41} \approx 6.40\,\text{A}, \quad \text{pf} = 5/\sqrt{41} \approx 0.781 \text{ lagging}.$$

Answer: $\boxed{|I| \approx 6.40\,\text{A}, \quad \text{pf} \approx 0.781 \text{ lagging}}$.

10.6 Extract resistance from a given admittance

Given $Y = G + \mathrm{j}B$, always invert before identifying the equivalent *series* resistance:

$$Z = \frac{G - \mathrm{j}B}{G^2 + B^2}, \qquad \boxed{R_{series} = \frac{G}{G^2 + B^2}}.$$

By contrast, $R_{parallel} = 1/G$ describes the shunt resistor in an equivalent parallel representation.

Worked example **Admittance-to-impedance**

$Y = 3 + \mathrm{j}4$ S. Then

$$Z = \frac{1}{3 + \mathrm{j}4} = \frac{3 - \mathrm{j}4}{25} = 0.12 - \mathrm{j}0.16\,\Omega.$$

Answer: $\boxed{R_{series} = 0.12\,\Omega}$.

10.7 Complex arithmetic shortcuts

For $Z_1 = R_1 + \mathrm{j}X_1$, $Z_2 = R_2 + \mathrm{j}X_2$,

$$Z_{series} = Z_1 + Z_2, \quad Z_{parallel} = \frac{Z_1 Z_2}{Z_1 + Z_2}.$$

Complex conjugate $(a + \mathrm{j}b)^* = a - \mathrm{j}b$ and $\mathrm{j}^2 = -1$. Rectangular coordinates simplify addition; polar coordinates simplify multiplication, division, and raising powers.

SSC JE trap / important distinction

A parallel AC branch with 3 A resistive current and 4 A inductive current does *not* draw 7 A total. Its total is $\sqrt{3^2 + 4^2} = 5$ A when branch currents are at 90°.

UNIT 11

AC power, power factor and correction

11.1 Derivation of real and reactive power

Take $v(t) = \sqrt{2}V\cos\omega t$ and $i(t) = \sqrt{2}I\cos(\omega t - \phi)$ (RMS V, I). Product-to-sum gives

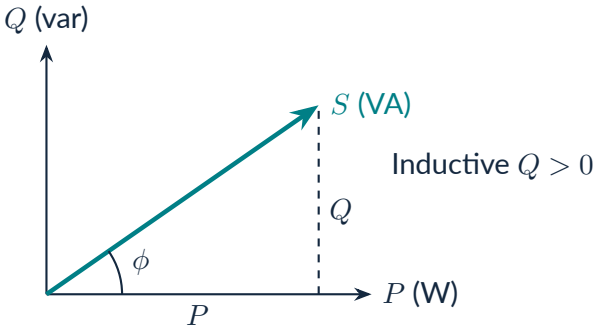
$$\begin{aligned} p(t) &= 2VI\cos\omega t\cos(\omega t - \phi) \\ &= VI\left[\cos\phi + \cos(2\omega t - \phi)\right]. \end{aligned}$$

The oscillating part averages to zero over a period. Consequently,

$$\boxed{P = VI\cos\phi\ \text{W}}, \quad \boxed{Q = VI\sin\phi\ \text{var}}, \quad \boxed{|S| = VI\ \text{VA}}.$$

Complex power convention (RMS phasors): $\underline{S} = \underline{V}\underline{I}^{*} = P + \mathrm{j}Q$. For passive *inductive* loads, $Q > 0$, and for passive *capacitive* loads, $Q < 0$.

$$|S| = \sqrt{P^2 + Q^2}, \qquad \tan\phi = \frac{Q}{P}, \qquad \text{pf} = \frac{P}{|S|} = \cos\phi.$$



Power triangle: S is the hypotenuse of P and Q.

Worked example — **Power triangle and power factor**

A load consumes $P = 4$ kW and $Q = 3$ kvar inductive.

$$|S| = \sqrt{4^2 + 3^2} = 5\,\text{kVA}, \quad \text{pf} = P/|S| = 4/5 = 0.8\ \text{lagging}.$$

Answer: $\boxed{|S| = 5\,\text{kVA},\ \text{pf} = 0.8\ \text{lagging}}$.

11.2 Relation to impedance

For a single equivalent series $R + \mathrm{j}X$, $P = I^2R$, $Q = I^2X$, and $|S| = I^2|Z|$. *Only* the resistive component dissipates average power in an ideal RLC circuit. The reactive components exchange stored energy with the source.

Worked example **Simple real power from rms ratings**

If $V = 230$ V RMS, $I = 5$ A RMS and pf $= 0.8$ lagging,

$$P = VI\cos\phi = 230(5)(0.8) = 920\,\text{W}.$$

Answer: $\boxed{P = 920\,\text{W}}$.

11.3 Lagging vs leading and disadvantages of low pf

Lagging pf normally corresponds to net inductive load and reactive current drawn from the source. Leading pf normally corresponds to net capacitive load. For fixed real power and voltage, $I = P/(V\,\text{pf})$: lowering pf increases current, resistive losses I^2R, voltage drop and equipment kVA rating. A correction capacitor reduces *net inductive vars* at the supply; it does not make the original load consume zero reactive power internally.

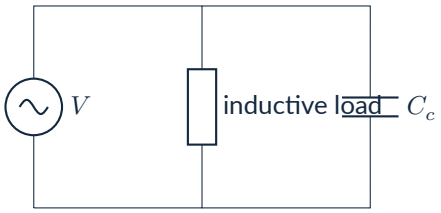
11.4 Power-factor correction, single phase

For fixed load real power P and change from lagging $\text{pf}_1 = \cos\phi_1$ to improved $\text{pf}_2 = \cos\phi_2$,

$$\boxed{Q_c = P(\tan\phi_1 - \tan\phi_2)}.$$

For a capacitor connected across a single-phase RMS source of voltage V,

$$Q_c = \omega C V^2 \quad \Longrightarrow \quad \boxed{C = \frac{Q_c}{\omega V^2}}.$$



Power-factor correction capacitor is in *shunt* across the inductive load.

Worked example **Single-phase capacitor sizing**

A 5 kW load at 230 V, 50 Hz improves from pf 0.6 lagging to 0.8 lagging.

$$\tan\phi_1 = \frac{0.8}{0.6} = \frac{4}{3}, \qquad \tan\phi_2 = \frac{0.6}{0.8} = \frac{3}{4}.$$

$$Q_c = 5000\left(\frac{4}{3} - \frac{3}{4}\right) = 2916.67\,\text{var}.$$

$$C = \frac{2916.67}{2\pi(50)(230)^2} \approx 175.5\,\mu\text{F}.$$

Answer: $\boxed{Q_c \approx 2.917\,\text{kvar}, \quad C \approx 175.5\,\mu\text{F}}$.

11.5 Power factor for non-sinusoidal waveforms

Under voltage/current distortion, *true* pf remains $P/(V_{rms}I_{rms})$ and need not equal the cosine of the fundamental phase displacement. If a problem involves harmonics, distinguish *displacement power factor* from *true power factor*. The common $\cos\phi$ shortcut assumes sinusoidal conditions.

SSC JE trap / important distinction

A reactive capacitor corrects inductive *source current*, but an excessive capacitor can over-correct into leading pf. For the same load P, source current is minimum at unity overall pf (under sinusoidal conditions).

UNIT 12

Resonance, quality factor and bandwidth

12.1 Series resonance: derive the condition

For series RLC, impedance $Z = R + \mathrm{j}(\omega L - 1/(\omega C))$. Resonance is the frequency at which net reactance vanishes:

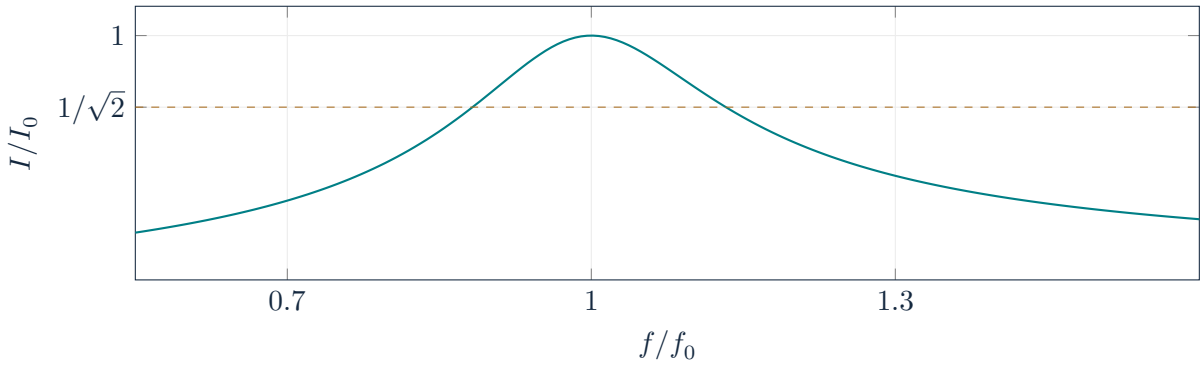
$$\omega_0 L = \frac{1}{\omega_0 C} \quad \Rightarrow \quad \boxed{\omega_0 = \frac{1}{\sqrt{LC}}}, \qquad \boxed{f_0 = \frac{1}{2\pi\sqrt{LC}}}.$$

At resonance: $Z = R$, current maximum $I_0 = V/R$ (for fixed applied sinusoidal voltage), pf unity, V_L and V_C equal in magnitude and opposite in phasor direction. Individually they may *greatly exceed* source voltage.

Key result

$$\boxed{Q_s = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R} = \frac{1}{R}\sqrt{\frac{L}{C}}}, \qquad \boxed{BW = f_2 - f_1 = \frac{R}{2\pi L} = \frac{f_0}{Q_s}}.$$

Here f_1, f_2 denote half-power frequencies for the *series-current* response: $|I| = I_0/\sqrt{2}$, average resistor power half its resonant value.



Illustrative series resonance response (quality factor determines curve width).

Worked example — **Resonance, quality and bandwidth**

A series circuit has $R = 10\,\Omega$, $L = 0.1$ H, $C = 100\,\mu$F.

$$\omega_0 = 1/\sqrt{0.1(10^{-4})} = 316.23\,\text{rad/s}; \qquad f_0 = 50.33\,\text{Hz}.$$

$$Q_s = \omega_0 L/R = 3.162, \qquad BW = R/(2\pi L) = 15.92\,\text{Hz}.$$

Answer: $\boxed{f_0 \approx 50.33\,\text{Hz},\ Q_s \approx 3.16,\ BW \approx 15.92\,\text{Hz}}$.

12.2 Voltage magnification in series resonance

At f_0, $V_L = IX_L = (V/R)\omega_0 L = Q_s V$ and likewise $V_C = Q_s V$ in magnitude. Thus

$$\boxed{V_L = V_C = Q_s V \quad \text{(magnitudes at series resonance)}}.$$

This does *not* violate KVL: phasor voltages across L and C cancel.

Worked example — **High component voltage despite modest source**

A series RLC has $R = 5\,\Omega$, $L = 400$ H, $C = 4$ F and is supplied by 400 V at its resonance frequency.

$$\omega_0 = \frac{1}{\sqrt{400 \cdot 4}} = \frac{1}{40}\ \text{rad/s}; \qquad X_C = \frac{1}{\omega_0 C} = 10\,\Omega.$$

$$I = 400/5 = 80\,\text{A}; \qquad V_C = 80(10) = 800\,\text{V}.$$

Answer: $\boxed{|V_C| = 800\,\text{V}}$.

12.3 Half-power frequencies: exact relation

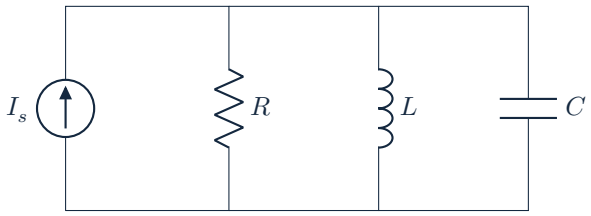
At $|X_L - X_C| = R$, $\omega L - 1/(\omega C) = \pm R$. The positive roots are

$$\omega_1 = \sqrt{\omega_0^2 + \left(\frac{R}{2L}\right)^2} - \frac{R}{2L}, \quad \omega_2 = \sqrt{\omega_0^2 + \left(\frac{R}{2L}\right)^2} + \frac{R}{2L}.$$

Their difference is exact for series RLC: $\omega_2 - \omega_1 = R/L$; geometric mean $\sqrt{\omega_1\omega_2} = \omega_0$. The shorthand $f_1 \approx f_0 - BW/2$, $f_2 \approx f_0 + BW/2$ requires high Q to be accurate.

12.4 Parallel resonance / anti-resonance

For an *ideal* parallel LC, the admittance is $Y = \text{j}(\omega C - 1/(\omega L))$. At $\omega_0 = 1/\sqrt{LC}$ the reactive branches exchange equal and opposite current; net external susceptance is zero. With a parallel resistor R present, source current is minimum (V/R) and input impedance is maximum (R). The ideal LC without any conductance has mathematically infinite input impedance at its resonant frequency.



Ideal parallel RLC: source current is minimum at susceptance cancellation.

12.5 Practical coil in parallel with capacitor

An actual inductor often has series coil resistance r, so its branch admittance is $1/(r + \text{j}\omega L)$. If the capacitor is ideal and r is frequency-independent,

$$\text{Im}\left\{\frac{1}{r + \text{j}\omega L} + \text{j}\omega C\right\} = 0 \ \Rightarrow \quad \boxed{\omega_r^2 = \frac{1}{LC} - \frac{r^2}{L^2}}$$

when the expression is positive. It does *not* generally resonate at exactly $1/\sqrt{LC}$, unlike the ideal case.

Worked example **A practical parallel branch**

An $L = 0.2$ H coil has series $r = 5\,\Omega$, connected in parallel with $C = 100\,\mu$F.

$$\omega_r = \sqrt{\frac{1}{(0.2)(10^{-4})} - \frac{5^2}{0.2^2}} = \sqrt{50000 - 625} = 222.2\,\text{rad/s}.$$

$$f_r = \omega_r/(2\pi) \approx 35.36\,\text{Hz}.$$

Answer: $\boxed{f_r \approx 35.36\,\text{Hz}}$.

12.6 How resonance changes with parameters

Since $f_0 \propto 1/\sqrt{LC}$, doubling L alone divides resonance frequency by $\sqrt{2}$; doubling both L and C divides it by 2; halving C multiplies it by $\sqrt{2}$. For fixed L, C, changing ideal series R changes Q and bandwidth but not f_0.

SSC JE trap / important distinction

At *series* resonance, source current is maximum. At *parallel* resonance with fixed applied voltage, source current is minimum (ideal high-impedance anti-resonance). Do not reverse the two.

UNIT 13

RC and RL switching transients (supporting practice)

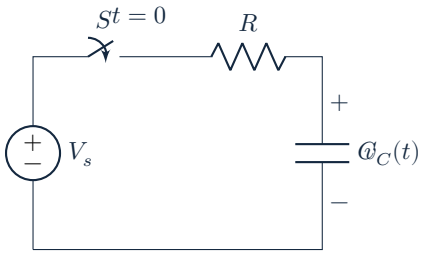
13.1 Why the final-value shortcut works

An ideal capacitor voltage cannot jump across a finite-current switching event; ideal inductor current cannot jump across finite-voltage switching. First-order networks obey

$$\boxed{x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}}, \quad t \geq 0$$

where x is a capacitor voltage or inductor current. The system is solved in three steps: identify initial state, find final DC state, calculate time constant.

13.2 RC charging/discharging



Series RC step circuit; continuity is on capacitor voltage.

KVL: $V_s = Ri + v_C = RC\,dv_C/dt + v_C$. Time constant $\tau = R_{th}C$ where R_{th} is the resistance seen by the capacitor with independent sources deactivated (under normal finite source models).

$$\boxed{v_C(t) = V_s + [v_C(0^+) - V_s]e^{-t/(RC)}}.$$

For initially uncharged capacitor, $v_C(t) = V_s(1 - e^{-t/RC})$, and $i(t) = (V_s/R)e^{-t/RC}$.

Worked example — **RC charging at one time constant**

An initially uncharged $100\,\mu$F capacitor is switched to a 10 V supply through $10\,\text{k}\Omega$.

$$\tau = RC = 10000(100 \times 10^{-6}) = 1\,\text{s}.$$

$$v_C(1) = 10(1 - e^{-1}) \approx 6.32\,\text{V}, \quad i(1) = \frac{10}{10000}e^{-1} = 0.368\,\text{mA}.$$

Answer: $\boxed{\tau = 1\,\text{s},\ v_C(1) \approx 6.32\,\text{V}}$.

13.3 RL build-up/decay

For RL excitation $V_s = Ri + L\,di/dt$, time constant $\tau = L/R_{th}$.

$$\boxed{i_L(t) = \frac{V_s}{R} + \left[i_L(0^+) - \frac{V_s}{R}\right] e^{-tR/L}}.$$

Worked example **RL current after switching**

$R = 5\,\Omega$, $L = 2$ H, $V_s = 10$ V and $i_L(0) = 0$.

$$\tau = L/R = 0.4\,\text{s}, \quad i_L(t) = 2(1 - e^{-t/0.4})\,\text{A}.$$

At $t = \tau$, $i_L = 2(1 - e^{-1}) \approx 1.264$ A.
Answer: $\boxed{i_L(0.4\,\text{s}) \approx 1.264\,\text{A}}$.

13.4 Timing and energy facts

At $t = \tau$ a rising first-order response reaches 63.2% of total change and a decaying one falls to 36.8% of its original excess. At $t = 3\tau$ rise $\approx 95\%$, at $5\tau \approx 99.3\%$. These are approximations to asymptotic settling, not exact finite-time completion.

SSC JE trap / important distinction

At a switching instant, the capacitor is *not always* a short circuit and the inductor is *not always* an open circuit. Those approximations apply only to some zero-initial-condition configurations at $t = 0^+$; use the actual stored initial voltage/current.

UNIT 14

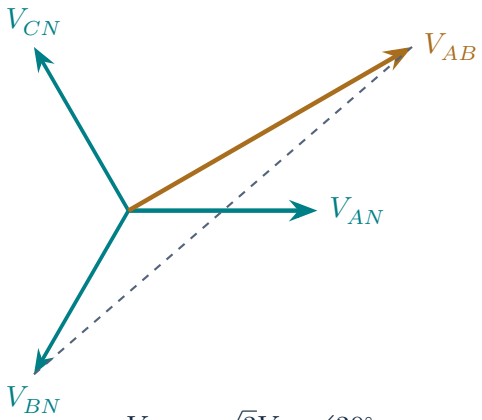
Balanced three-phase circuits and star–delta loads

14.1 Balanced supply: phase sequence and phasors

Three equal-magnitude sinusoidal phase EMFs displaced by 120° form a balanced three-phase supply. Positive-sequence phase (line-to-neutral) voltages may be written:

$$\underline{V}_{AN} = V_{ph}\angle 0^\circ, \quad \underline{V}_{BN} = V_{ph}\angle(-120^\circ), \quad \underline{V}_{CN} = V_{ph}\angle 120^\circ.$$

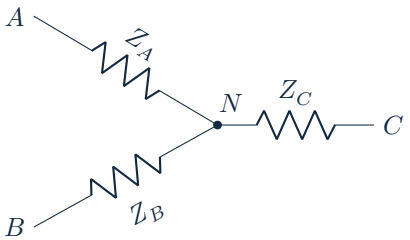
For a positive-sequence star connection, line voltage $\underline{V}_{AB} = \underline{V}_{AN} - \underline{V}_{BN}$ leads phase voltage $\underline{V}_{AN}$ by 30°. The corresponding magnitudes satisfy $V_L = \sqrt{3}V_{ph}$.



Positive sequence, $V_{AB} = \sqrt{3}V_{AN}\angle 30^\circ$

Phasor subtraction explains the $\sqrt{3}$ and 30° factors.

14.2 Star/Y load connections



Three load impedances have one common neutral point.

For balanced star,

$$\boxed{V_L = \sqrt{3}V_{ph}}, \qquad \boxed{I_L = I_{ph}}, \qquad I_{ph} = \frac{V_{ph}}{|Z_{ph}|}.$$

With balanced linear impedances, phase currents sum to zero, so neutral current is zero even when neutral is connected. In unbalanced star, neutral current generally need not be zero.

Worked example **Balanced star load**

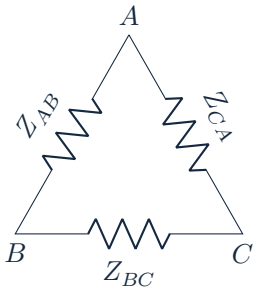
A 3-phase 400 V line-to-line source supplies three equal star phase impedances $Z_{ph} = 8 + \text{j}6\,\Omega$.

$$|Z_{ph}| = 10\,\Omega, \quad V_{ph} = 400/\sqrt{3} = 230.94\,\text{V}.$$

$$I_L = I_{ph} = 230.94/10 = 23.094\,\text{A}, \quad \text{pf} = 8/10 = 0.8 \text{ lagging}.$$

Answer: $\boxed{I_L \approx 23.09\,\text{A}, \quad \text{pf} = 0.8 \text{ lagging}}$.

14.3 Delta-connected load connections



Delta loads are between pairs of lines, with no common neutral point.

For balanced delta,

$$\boxed{V_L = V_{ph}}, \qquad \boxed{I_L = \sqrt{3} I_{ph}}, \qquad I_{ph} = \frac{V_L}{|Z_{ph}|}.$$

For positive sequence, a line current (e.g. $I_A = I_{AB} - I_{CA}$) lags its corresponding delta phase current I_{AB} by 30°. This is independent of the load pf angle, which still enters the phase current relative to its own phase voltage.

Worked example **Balanced delta load**

A balanced delta of $Z_{ph} = 12 + \text{j}16\,\Omega$ is connected to 400 V line-to-line.

$$|Z| = 20\,\Omega, \quad I_{ph} = 400/20 = 20\,\text{A}, \quad I_L = \sqrt{3}(20) = 34.64\,\text{A}.$$

Answer: $\boxed{I_L \approx 34.64\,\text{A}, \quad \text{pf} = 0.6 \text{ lagging}}$.

14.4 One-page star–delta comparison

Quantity	Balanced star (Y)	Balanced delta (Δ)
Phase voltage	$V_{ph} = V_L/\sqrt{3}$	$V_{ph} = V_L$
Phase current	$I_{ph} = I_L$	$I_{ph} = I_L/\sqrt{3}$
Line current	$I_L = I_{ph}$	$I_L = \sqrt{3} I_{ph}$
Balanced impedance conversion	$Z_Y = Z_\Delta/3$	$Z_\Delta = 3Z_Y$
3-phase total real power	$\sqrt{3} V_L I_L \cos\phi$	$\sqrt{3} V_L I_L \cos\phi$

14.5 Three-phase power from phase and line quantities

For a balanced load, real, reactive and apparent powers are:

$$\boxed{P = 3V_{ph}I_{ph}\cos\phi = \sqrt{3}V_L I_L \cos\phi},$$

$$\boxed{Q = \sqrt{3}V_L I_L \sin\phi}, \qquad \boxed{|S| = \sqrt{3}V_L I_L}.$$

These assume rms line values, balanced sinusoidal phasors and the phase voltage-to-phase current angle ϕ. For unbalanced loads, compute $\sum V_{ph,k}I^*_{ph,k}$ using individual phase quantities.

Worked example **Line current from three-phase power**

A balanced 15 kW, 400 V (line) motor has pf 0.8 lagging. Find source line current ignoring internal load details.

$$I_L = \frac{15000}{\sqrt{3}(400)(0.8)} \approx 27.06\,\text{A}.$$

Answer: $\boxed{I_L \approx 27.1\,\text{A}}$.

14.6 Balanced load conversion: why impedance /3

At the same three line terminals, an equal delta with branch impedance Z_Δ is externally equivalent to a star with $Z_Y = Z_\Delta/3$. You may replace delta by star to simplify a problem; when doing so, retain the same *line voltage*. Individual branch voltages and currents change, but source line currents and total power remain the same.

Worked example **Delta-to-star three-phase equivalent**

Each delta phase impedance is $45\,\Omega$ (resistive). The equivalent balanced star phase resistance is

$$R_Y = R_\Delta/3 = 45/3 = 15\,\Omega.$$

Answer: $\boxed{R_Y = 15\,\Omega}$.

14.7 Reconnecting the same three impedances

For the *same three equal resistances* with the same line voltage V_L, delta connection gives phase voltage V_L and star gives $V_L/\sqrt{3}$. Thus

$$P_\Delta = 3\frac{V_L^2}{R}, \qquad P_Y = 3\frac{(V_L/\sqrt{3})^2}{R} = \frac{V_L^2}{R}.$$

Therefore $\boxed{P_\Delta = 3P_Y}$ for purely resistive branches or for identical constant impedances when comparing real power at unchanged supply voltage.

14.8 Power factor correction, balanced three-phase

If P is the entire three-phase real load power, the required total compensating vars remain $Q_c = P(\tan\phi_1 - \tan\phi_2)$. With C *per phase*:

$$\boxed{C_\Delta = \frac{Q_c}{3\omega V_L^2}}, \qquad \boxed{C_Y = \frac{Q_c}{\omega V_L^2}}.$$

The delta-connected capacitor needs only one-third the per-branch capacitance of a star bank to deliver the same total vars at the same line voltage (ideal balanced conditions).

14.9 Two-wattmeter method

A standard three-wire two-wattmeter measurement gives *total real power* $P = W_1 + W_2$ even for unbalanced loads under appropriate connections. For balanced sinusoidal loads only, one common labeling yields

$$W_1 = V_L I_L \cos(30^\circ - \phi), \quad W_2 = V_L I_L \cos(30^\circ + \phi),$$
$$\tan\phi = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}, \qquad \text{(consistent labeling required).}$$

At pf 0.5 ($\phi = 60^\circ$) one wattmeter reads zero; below pf 0.5 one reading is negative. Swap voltage-coil polarity physically to obtain upscale indication if needed, then keep the algebraic negative sign when summing.

Worked example **Two-wattmeter power factor**

A balanced load records $W_1 = 6$ kW, $W_2 = 2$ kW in a convention where W_1 is the larger reading for a lagging load.

$$\tan\phi = \sqrt{3}\frac{6-2}{6+2} = \frac{\sqrt{3}}{2} = 0.8660 \quad \Rightarrow \quad \text{pf} = \frac{1}{\sqrt{1+0.866^2}} \approx 0.756.$$

Answer: $\boxed{P = 8\,\text{kW}, \quad \text{pf} \approx 0.756 \text{ lagging}}$.

SSC JE trap / important distinction

For Y use $V_L = \sqrt{3}V_{ph}$; for Δ use $I_L = \sqrt{3}I_{ph}$. A common SSC error is multiplying by $\sqrt{3}$ twice in the same circuit.

UNIT 15

Advanced mixed models and exam-solving strategy

15.1 How to select the correct method quickly

Problem clue	First method to try
Circuit contains obvious series/parallel sections	Reduce them; use divider ratios
Only one or two node potentials needed	Nodal KCL and supernode for voltage sources
Few planar loops; mostly voltage sources	Mesh KVL and supermesh for current sources
Load varies or asked maximum load power	Thevenin or Norton at load terminals
Several independent sources / asked contribution	Superposition; combine voltages/currents, then power
Unknown bridge current; ratios suggest balance	Check Wheatstone balance or symmetry
Coils in aiding/opposing, k or M requested	$L_+ - L_- = 4M$ and $M = k\sqrt{L_1 L_2}$
AC series resistors/coils/caps	Add impedances, find $\lvert Z \rvert$ and angle
AC parallel branches	Add admittances/phasor currents
Resonance / bandwidth / voltage magnification	Set $X_L = X_C$, then compute Q, BW
Three-phase line quantities	Identify Y/Δ before using $\sqrt{3}$

15.2 Model A: a resistor shorted by ideal wire

If an ideal wire connects the same two terminals as a resistor, the voltage across the resistor is zero and current through that resistor is 0 for nonzero R. This is distinct from a resistor in *series* with an open circuit: then entire path current is zero but other node voltages depend on topology.

Worked example **Shorted branch**

A $10\,\Omega$ resistor is directly paralleled by an ideal $0\,\Omega$ wire. A total 5 A enters the two-branch combination. Equivalent resistance $0\,\Omega$; the resistor drop is zero, so its current is zero. The wire carries 5 A.

Answer: $\boxed{I_{10\Omega} = 0, \quad I_{\text{wire}} = 5\,\text{A}}$.

15.3 Model B: converting a desired resistor power into source current

If a branch resistor dissipates 36 W and has $R = 9\,\Omega$, first find branch current $I = \sqrt{P/R} = 2$ A, not P/R. Then use branch KCL or voltage divider to infer source current.

Worked example **Known resistor power**

A $9\,\Omega$ resistor and a $6\,\Omega$ resistor are in parallel and the first dissipates 36 W.

$$V = \sqrt{PR} = \sqrt{36(9)} = 18\,\text{V}, \quad I_9 = 2\,\text{A}, \quad I_6 = 3\,\text{A}.$$

Answer: $\boxed{I_{total} = 5\,\text{A}}$.

15.4 Model C: a source current division and terminal voltage

A 12 A source drives $3\,\Omega$ and $6\,\Omega$ shunt resistors. $R_{eq} = 2\,\Omega$ and common voltage is $V = 12 \times 2 = 24$ V. Branch currents are 8 and 4 A, adding to 12 A. This is an easy check against accidental inverse-current-division errors.

15.5 Model D: compound RL/RC phasors

Worked example **Power factor from parallel branch powers**

A 100 V source feeds a resistive branch $R = 20\,\Omega$ and capacitive branch $X_C = 25\,\Omega$ in parallel. Then $I_R = 5$ A, $I_C = +\text{j}4$ A.

$$I_{total} = 5 + \text{j}4, \qquad |I| = \sqrt{41} = 6.403\,\text{A}, \qquad \text{pf} = 5/\sqrt{41} = 0.781.$$

Answer: $\boxed{\text{pf} = 0.781 \text{ leading}}$.

15.6 Model E: line voltage ambiguity

Three-phase questions often state “400 V supply” without restating the reference. Unless a phase voltage is explicitly indicated, a three-phase supply rating usually means *line-to-line RMS*. Use given wording rather than assuming all supplies are star connected.

Worked example **Power for fixed branch impedance under reconnection**

Three $30\,\Omega$ identical resistors are first star connected, then delta connected, on the same 300 V *line* supply.

$$P_Y = \frac{300^2}{30} = 3000\,\text{W}; \qquad P_\Delta = 3\frac{300^2}{30} = 9000\,\text{W}.$$

Answer: $\boxed{P_Y = 3\,\text{kW},\ P_\Delta = 9\,\text{kW}}$.

15.7 Model F: power maximum versus power efficiency

Worked example **Maximum transfer does not mean high efficiency**

For $V_{th} = 20$ V and $R_{th} = 4\,\Omega$, optimum resistive $R_L = 4\,\Omega$.

$$I = 20/(4+4) = 2.5\,\text{A}; \quad P_L = 2.5^2(4) = 25\,\text{W}; \quad \eta = \frac{4}{4+4} = 50\%.$$

Answer: $\boxed{P_{max} = 25\,\text{W}, \quad \eta = 50\%}$.

15.8 Model G: question asks resistance from admittance

Worked example **Invert complex admittance first**

Given $Y = 6 - \text{j}8$ S, $|Y| = 10$ S. Since $Z = (6+\text{j}8)/100 = 0.06 + \text{j}0.08\,\Omega$, the *series* resistance is $0.06\,\Omega$. A branch model might have parallel resistance $1/6\,\Omega$ instead.
Answer: $\boxed{R_{series} = 0.06\,\Omega}$.

15.9 Final 20-second answer audit

Check that (1) units have been converted, (2) the chosen quantity is RMS or peak, (3) series/parallel conditions are used correctly, (4) negative imaginary reactance is capacitive, (5) any independent sources are deactivated correctly, (6) three-phase line and phase variables are not mixed, and (7) final answer order of magnitude makes sense.

Key result

Fast numerical identities:
3–4–5 triangle: $\sqrt{3^2+4^2} = 5$; 6–8–10 similarly. $\cos 36.87^\circ = 0.8$, $\sin 36.87^\circ = 0.6$, $\tan 36.87^\circ = 0.75$.
$\sqrt{3} \approx 1.732$, $1/\sqrt{3} \approx 0.577$, $\sqrt{2} \approx 1.414$, $\pi \approx 3.142$.

UNIT 16

Authentic previous-year-question review, 2023–2025

Key result

How to use this chapter. Every numbered item below is keyed to a distinct question in a publicly circulated SSC JE Electrical paper or transcript (year, paper, shift, question number). The question wording is *paraphrased* to focus on the electrical model rather than reproducing an entire paper. Solutions are independently derived, not copied from an answer key. The 2023 and 2024 examples are traced to complete question-paper PDFs; the 2025 examples come from a public transcription, *not* a verified official SSC-hosted question-paper PDF. Details and direct source links are at the back.

16.1 2025 Paper I: publicly transcribed December 4, Shift 2

These are *2025-paper-attributed* items. The public transcript was reviewed for question text, and each numerical result was checked from first principles. Use these for pattern recognition; for legal/exact wording or official key disputes rely on SSC's candidate-response record.

Worked example — **2025 Dec 4 Shift 2, Q101 — coupled inductors**

Two series-connected coupled windings have equivalent inductances of 150 mH aiding and 80 mH opposing. Find the mutual inductance.
Recognize: both winding orientations are given. Form the difference to eliminate $L_1 + L_2$.

$$L_+ = L_1 + L_2 + 2M, \qquad L_- = L_1 + L_2 - 2M; \quad M = \frac{150 - 80}{4} = 17.5\,\text{mH}.$$

Answer: $\boxed{M = 17.5\,\text{mH}}$.

SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q101; public paper transcription

Worked example — **2025 Dec 4 Shift 2, Q102 — resistor colour bands**

A four-band resistor reads red, red, orange and gold. Red=2, orange multiplier 10^3, gold tolerance $\pm 5\%$.

$$R = 22 \times 10^3\,\Omega = 22\,\text{k}\Omega.$$

Answer: $\boxed{R = 22\,\text{k}\Omega \ \pm 5\%}$.

SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q102; public paper transcription

Worked example — **2025 Dec 4 Shift 2, Q104 — practical battery**

A 12 V battery has $0.1\,\Omega$ internal resistance and supplies $1.9\,\Omega$ external load. Calculate terminal voltage.

$$I = \frac{12}{1.9 + 0.1} = 6\,\text{A}, \qquad V_t = E - Ir = 12 - 6(0.1) = 11.4\,\text{V}.$$

Variant: during charging, terminal voltage is $V_t = E + Ir$ with the consistent charging current direction.
Answer: $\boxed{V_t = 11.4\,\text{V}}$.
SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q104; public paper transcription

Worked example **2025 Dec 4 Shift 2, Q105 — current division**

A total current of 10 A enters parallel $4\,\Omega$ and $6\,\Omega$ resistors. What flows through the $6\,\Omega$ branch?

$$I_6 = 10\frac{4}{4+6} = 4\,\text{A}.$$

Trap: current is inversely proportional to resistance; the $6\,\Omega$ branch takes less than half.
Answer: $\boxed{I_6 = 4\,\text{A}}$.
SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q105; public paper transcription

Worked example **2025 Dec 4 Shift 2, Q106 — electrical energy**

A 240 V load draws 5 A continuously for 30 minutes. Express the energy in joules.

$$t = 30 \times 60 = 1800\,\text{s}, \quad W = VIt = 240(5)(1800) = 2{,}160{,}000\,\text{J}.$$

Answer: $\boxed{W = 2.16\,\text{MJ} = 0.6\,\text{kWh}}$.
SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q106; public paper transcription

Worked example **2025 Dec 4 Shift 2, Q141–143 — magnetic definitions**

Three neighboring conceptual items test whether reluctance $\mathcal{R}$, flux Φ, magnetomotive force $\mathcal{F}$, flux density B, and field intensity H are distinguished.

$$\boxed{\mathcal{F} = \Phi\mathcal{R} = NI}, \qquad \boxed{L = \frac{N\Phi}{I}}, \qquad \boxed{\mu = \frac{B}{H}}.$$

The middle relation assumes flux Φ links all N turns, and the last relation applies to a linear isotropic region or its local effective permeability.
Answer: $\boxed{\mathcal{F} = \Phi\mathcal{R},\ L = N\Phi/I,\ \mu = B/H}$.
SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q141,Q142,Q143; public transcription

Worked example **2025 Dec 4 Shift 2, Q149 — Norton to Thevenin**

The Norton network has source current 5 A in parallel with $4\,\Omega$. What Thevenin voltage does it represent?

$$V_{th} = I_N R_N = 5 \times 4 = 20\,\text{V}.$$

The equivalent Thevenin resistance remains $4\,\Omega$; only the arrangement changes.
Answer: $\boxed{V_{th} = 20\,\text{V}}$.
SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q149; public paper transcription

Worked example **2025 Dec 4 Shift 2, Q151–153 — equivalences and balanced delta**

Three conceptual items in sequence ask for a Norton arrangement, ideal voltage-source internal resistance, and balanced Δ to Y impedance conversion.

$$\text{Norton: ideal current source} \parallel R_N, \quad R_{\text{ideal voltage source}} = 0, \quad Z_Y = \frac{Z_\Delta}{3} = \frac{45}{3} = 15\,\Omega.$$

For the impedance conversion, the $45\,\Omega$ is *each delta side*, and the $15\,\Omega$ is *each corresponding star branch*.

Answer: $\boxed{Z_Y = 15\,\Omega}$.

SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q151,Q152,Q153; public transcription

Worked example **2025 Dec 4 Shift 2, Q171 — toroidal magnetic field**

A 300-turn toroid has mean magnetic path length 50 cm, and current 2.5 A. Find the magnetic field intensity (ideal uniform model).

$$\ell = 0.50\,\text{m}, \quad \mathcal{F} = NI = 750\,\text{At}, \qquad H = \frac{NI}{\ell} = \frac{750}{0.5} = 1500\,\text{A/m}.$$

Answer: $\boxed{H = 1500\,\text{A/m}}$.

SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q171; public paper transcription

Worked example **2025 Dec 4 Shift 2, Q172 — solenoid inductance**

A coil has $N = 400$ turns, $\mu_r = 1500$, area $5\,\text{cm}^2$ and magnetic path 30 cm. Determine inductance neglecting leakage.

$$A = 5 \times 10^{-4}\,\text{m}^2, \quad \ell = 0.30\,\text{m},$$

$$L = \frac{\mu_0 \mu_r N^2 A}{\ell} = \frac{(4\pi \times 10^{-7})(1500)(400)^2(5 \times 10^{-4})}{0.30} \approx 0.503\,\text{H}.$$

Answer: $\boxed{L \approx 0.503\,\text{H}}$.

SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q172; public paper transcription

Worked example **2025 Dec 4 Shift 2, Q187 — line charging-capacitance model**

A line has capacitance 10 nF per km over 150 km. At 50 Hz, what is the magnitude of equivalent capacitive reactance (lumped approximation)?

$$C = 150(10 \times 10^{-9}) = 1.5 \times 10^{-6}\,\text{F}, \quad X_C = \frac{1}{2\pi(50)(1.5 \times 10^{-6})} \approx 2122\,\Omega.$$

Answer: $\boxed{X_C \approx 2.12\,\text{k}\Omega}$.

SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q187; public paper transcription

Worked example **2025 Dec 4 Shift 2, Q188–189 — single-phase real power**

Two nearby questions test the same power-factor formula in opposite directions. For a 10 kW, 230 V single-phase load at 0.85 power factor,

$$I = \frac{P}{V \cos\phi} = \frac{10000}{230(0.85)} \approx 51.15\,\text{A}.$$

For 230 V, 5 A, 0.8 pf,

$$P = VI\cos\phi = 230(5)(0.8) = 920\,\text{W}.$$

Answer: $\boxed{I \approx 51.2\,\text{A}, \quad P = 920\,\text{W}}$.

SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q188,Q189; public transcription

Worked example **2025 Dec 4 Shift 2, Q192–195 — rapid fundamental group**

These consecutive basics test four key results: $6\,\Omega \parallel 3\,\Omega = 2\,\Omega$; KVL means algebraic voltage rises and drops around a loop sum to zero; maximum power to a variable *DC resistance* occurs at $R_L = R_{th}$; and a 10 V open-circuit, $5\,\Omega$ Thevenin network is a 10 V ideal source in series with $5\,\Omega$.

$$R_{eq} = \frac{6\cdot 3}{6+3} = 2\,\Omega, \qquad P_{\max} = \frac{V_{th}^2}{4R_{th}}.$$

Answer: $\boxed{R_{eq} = 2\,\Omega, \quad R_L = R_{th}}$.

SOURCE IDENTIFIER: 2025-P1-04DEC-S2-Q192,Q193,Q194,Q195; public transcription

16.2 2024 Paper I and Paper II: traced question-paper PDFs

Worked example **2024 June 5, Paper I Electrical, Shift 3, Q14 — capacitor current**

An ideal 10 F capacitor has $v(t) = 10t^3 - 5t + 10$ volts. Find instantaneous current at $t = 2$ seconds.

$$i = C\frac{dv}{dt} = 10(30t^2 - 5), \qquad i(2) = 10(120 - 5) = 1150\,\text{A}.$$

Recognize: an instantaneous voltage *function* is given; differentiation precedes substitution.
Answer: $\boxed{i(2) = 1150\,\text{A}}$.

SOURCE IDENTIFIER: 2024-P1-05JUN-S3-Q14; complete reproduced question-paper PDF

Worked example **2024 June 5, Paper I Electrical, Shift 3, Q15 — resonant capacitor voltage**

A series circuit has $R = 5\,\Omega$, $L = 400$ H, $C = 4$ F and applied voltage 400 V RMS. Find capacitor voltage at resonance.

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1600}} = 0.025\,\text{rad/s}, \quad X_C = \frac{1}{\omega_0 C} = 10\,\Omega.$$

$$I_0 = \frac{V}{R} = \frac{400}{5} = 80\,\text{A}, \qquad V_C = I_0 X_C = 80(10) = 800\,\text{V}.$$

Crucial: V_L or V_C can exceed applied voltage, despite their *phasor sum* cancelling. A reproduction may show a previous candidate's *chosen option*; do not mistake that label for the correct response.
Answer: $\boxed{V_C = 800\,\text{V}}$.

SOURCE IDENTIFIER: 2024-P1-05JUN-S3-Q15; independently recalculated

Worked example **2024 June 5, Paper I Electrical, Shift 3, Q78 — harmonics RMS**

Given $v(t) = 2\sin\omega t + 3\sin 3\omega t + 5\sin 5\omega t$ volts, find total RMS voltage.

$$V_{rms} = \sqrt{\frac{2^2 + 3^2 + 5^2}{2}} = \sqrt{19} \approx 4.359\,\text{V}.$$

Reason: orthogonal harmonics add by *mean squares*, not by peak amplitudes.

Answer: $\boxed{V_{rms} = \sqrt{19}\,\text{V}}$.

SOURCE IDENTIFIER: 2024-P1-05JUN-S3-Q78; complete question-paper PDF

Worked example **2024 June 6, Paper I Electrical, Shift 2, Q40 — superposition scope**

Determine where superposition applies in electrical circuit analysis. The operational rule is to sum *currents or voltages* produced by individual independent sources in a *linear* network; do not directly sum powers.

$$i(t) = i_1(t) + i_2(t) + \cdots, \qquad P_R = R(i_1 + i_2 + \cdots)^2.$$

The cross terms in P_R explain why power itself is generally not superposable.

SOURCE IDENTIFIER: 2024-P1-06JUN-S2-Q40; complete question-paper PDF

Worked example **2024 June 6, Paper I Electrical, Shift 2, Q83 — KVL**

A conceptual item tests which circuit law equates the algebraic sum of potential differences around a closed path to zero.

$$\boxed{\sum_{\text{closed loop}} v_k = 0.}$$

Recognize: a *loop* points to KVL; a *junction* points to KCL.

SOURCE IDENTIFIER: 2024-P1-06JUN-S2-Q83; complete question-paper PDF

Worked example **2024 June 7, Paper I Electrical, Shift 3, Q44 — aiding and opposing**

The measured series inductance of a coupled pair is 38 mH for aiding and 12 mH for opposing connection. Determine M.

$$4M = L_+ - L_- = 38 - 12 = 26\,\text{mH}, \quad M = 6.5\,\text{mH}.$$

Answer: $\boxed{M = 6.5\,\text{mH}}$.

SOURCE IDENTIFIER: 2024-P1-07JUN-S3-Q44; complete question-paper PDF

Worked example **2024 June 7, Paper I Electrical, Shift 3, Q86 — air-gap fringing**

An air gap in a magnetic path causes flux lines to spread outward into a larger effective area. This is *fringing*. Compared with assuming perfectly parallel flux lines confined to the core area, fringing increases effective gap area and reduces the corresponding idealized gap reluctance.

$$\mathcal{R}_g \approx \frac{g}{\mu_0 A_{\text{effective}}}.$$

Answer: $\boxed{\text{air-gap fringing}}$.

SOURCE IDENTIFIER: 2024-P1-07JUN-S3-Q86; complete question-paper PDF

Worked example **2024 June 7, Paper I Electrical, Shift 3, Q87 — upper bound on mutual inductance**

Two coupled inductors have self-inductances 20 H and 320 H. Find maximum physically allowed mutual inductance.

$$|M| \le \sqrt{L_1 L_2} = \sqrt{20(320)} = \sqrt{6400} = 80\,\text{H}.$$

Equality requires perfect coupling, $|k| = 1$.

Answer: $\boxed{M_{\max} = 80\,\text{H}}$.

SOURCE IDENTIFIER: 2024-P1-07JUN-S3-Q87; complete question-paper PDF

Worked example **2024 November 6, Paper II Electrical, Q16 — Thevenin load current**

The network at a load has $V_{th} = 40$ V and $R_{th} = 5\,\Omega$; a $15\,\Omega$ load is connected. Find current.

$$I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{40}{5 + 15} = 2\,\text{A}.$$

Trap: the load is *series* with Thevenin resistance, not parallel with it.

Answer: $\boxed{I_L = 2\,\text{A}}$.

SOURCE IDENTIFIER: 2024-P2-06NOV-Q16; complete reproduced question-paper PDF

16.3 2023 Paper I: traced October sessions

Worked example **2023 October 11, Shift 3, Q20 — R–L impedance**

A resistor and inductor are in series across an AC voltage source. Find impedance magnitude in terms of resistance R, inductance L, and angular frequency ω.

$$Z = R + \text{j}\omega L, \qquad \boxed{|Z| = \sqrt{R^2 + (\omega L)^2}}.$$

Trap: do not add $R + \omega L$ as ordinary scalars because resistance and inductive reactance lie on perpendicular phasor axes.

SOURCE IDENTIFIER: 2023-P1-11OCT-S3-Q20; reproduced full paper

Worked example **2023 October 11, Shift 3, Q23 — inductor stored energy**

A 4 H inductor carries 15 A. Find stored magnetic energy.

$$W = \frac{1}{2}LI^2 = \frac{1}{2}(4)(15)^2 = 450\,\text{J}.$$

Answer: $\boxed{W = 450\,\text{J}}$.

SOURCE IDENTIFIER: 2023-P1-11OCT-S3-Q23; reproduced full paper

Worked example **2023 October 11, Shift 3, Q80 — RMS from peak**

A sinusoid has peak value 388 V. What is its RMS value?

$$V_{rms} = \frac{388}{\sqrt{2}} \approx 274.36\,\text{V}.$$

For exam options rounded to integers, 275 V is the closest common choice.

Answer: $\boxed{V_{rms} \approx 274.4\,\text{V}}$.
SOURCE IDENTIFIER: 2023-P1-11OCT-S3-Q80; reproduced full paper

Worked example **2023 October 11, Shift 3, Q81 — series mutual inductance**

Series aiding and opposing values for a coupled pair are 4 H and 2 H. Find their mutual inductance.

$$M = \frac{L_+ - L_-}{4} = \frac{4-2}{4} = 0.5\,\text{H}.$$

Answer: $\boxed{M = 0.5\,\text{H}}$.
SOURCE IDENTIFIER: 2023-P1-11OCT-S3-Q81; reproduced full paper

Worked example **2023 October 11, Shift 3, Q82 — safe series resistance**

An LED with 3 V forward drop is connected to a 12 V DC source; desired current is 36 mA. What series resistance is required? (Electronic fundamentals cross-topic.)

$$R = \frac{12-3}{0.036} = 250\,\Omega.$$

Answer: $\boxed{R = 250\,\Omega}$.
SOURCE IDENTIFIER: 2023-P1-11OCT-S3-Q82; reproduced full paper

Worked example **2023 October 11, Shift 3, Q83 — parallel-plate capacitance**

A capacitor in air has area 200 cm^2, separation 10 cm; $\varepsilon_0 = 8.854\times10^{-12}$ F/m. Compute capacitance.

$$A = 200 \times 10^{-4} = 0.02\,\text{m}^2, \quad d = 0.10\,\text{m},$$

$$C = \frac{\varepsilon_0 A}{d} = \frac{8.854 \times 10^{-12}(0.02)}{0.1} = 1.7708 \times 10^{-12}\,\text{F}.$$

Answer: $\boxed{C \approx 1.77\,\text{pF}}$.
SOURCE IDENTIFIER: 2023-P1-11OCT-S3-Q83; reproduced full paper

Worked example **2023 October 9, Shift 3, Q5 — apparent power**

A single-phase load takes real power 40 W at pf 0.8. Find apparent power.

$$S = \frac{P}{\cos\phi} = \frac{40}{0.8} = 50\,\text{VA}.$$

Answer: $\boxed{S = 50\,\text{VA}}$.
SOURCE IDENTIFIER: 2023-P1-09OCT-S3-Q5; reproduced full paper

Worked example **2023 October 9, Shift 3, Q28 — series resistance from admittance**

A circuit has total admittance $Y = 3 + \text{j}4$ S. What is the *series equivalent resistance*?

$$Z = \frac{1}{3+\text{j}4} = \frac{3-\text{j}4}{25} = 0.12 - \text{j}0.16\,\Omega.$$

Check: $1/G = 1/3\,\Omega$ would be the parallel-branch resistance, *not* series resistance.
Answer: $\boxed{R_{\text{series}} = 0.12\,\Omega}$.

SOURCE IDENTIFIER: 2023-P1-09OCT-S3-Q28; reproduced full paper

16.4 What repeated across the reviewed PYQs?

Repeated skill	Evidence among samples	Next practice
Series–parallel and source models	2025 Q104/105/149/151–153/192–195; 2024 P2 Q16	Drills 1–20
Coupled coils and magnetics	2025 Q101/141–143/171/172; 2024 Q44/86/87; 2023 Q23/81	Drills 27–36
Capacitors and AC waveforms	2024 Q14/78; 2023 Q80/83	Drills 21–26, 37–39
AC power, impedance, resonance	2025 Q187–189; 2024 Q15; 2023 Q5/20/28	Drills 40–49

SSC JE trap / important distinction

Do not turn this into a false weightage chart. It is a *curated PYQ sample*, not a census of every electrical question from three years or a statistically validated topic-frequency forecast. It demonstrates recurring methods without estimating an official chapter-wise mark distribution.

UNIT 17

Original exam practice: 64 questions + explained answer key

Every question here is *original practice*, not a claimed SSC PYQ. Attempt the entire section *before* reading the explanations. For speed work, target 60–90 seconds for straightforward questions and 2–3 minutes for multistep questions.

17.1 Question bank

A. DC laws, resistive networks and theorems

1. A 12 V source is connected to $3\,\Omega$. Find the current.
2. A 48 V supply feeds $4\,\Omega$ and $12\,\Omega$ in series. Find current.
3. Find the parallel equivalent of $20\,\Omega$ and $30\,\Omega$.
4. $2\,\Omega$ and $4\,\Omega$ are in series across 18 V. Find voltage on $4\,\Omega$.
5. 5 A divides between $4\,\Omega$ and $6\,\Omega$ in parallel. Find current in $4\,\Omega$.
6. A uniform wire is stretched to twice its original length without losing material. Find R'/R.
7. A conductor is $10\,\Omega$ at 20°C and has $\alpha_{20} = 0.004/^\circ$C. Find R at 70°C.
8. A $12\,\Omega$ resistor dissipates 48 W. Find RMS or DC current.
9. At a node, 2 A enters and 1.2 A and 0.3 A leave. What additional current must leave?
10. A $5\,\Omega$ element connects a 10 V node to ground. Find current from the node to ground.
11. $V_{th} = 9$ V, $R_{th} = 3\,\Omega$ and $R_L = 6\,\Omega$. Find load current.
12. A 24 V Thevenin source has resistance $6\,\Omega$. Find maximum power to a resistive load.
13. A Norton network has $I_N = 4$ A and $R_N = 5\,\Omega$. Find open-circuit voltage.
14. Convert a 10 V ideal source series with $2\,\Omega$ into Norton form.
15. A Wheatstone bridge has arms $R_1 = 4, R_2 = 8, R_3 = 6\,\Omega$. Find balanced fourth arm R_4.
16. Convert a balanced Δ with each side $21\,\Omega$ to Y.
17. Convert balanced Y arms of $5\,\Omega$ each to a delta.
18. Total admittance is $Y = 4 + \mathrm{j}3$ S. Find series-equivalent real resistance.
19. Find power in a $100\,\Omega$ resistor across 200 V DC.
20. Find $15\,\Omega \parallel 30\,\Omega$.

B. Capacitors, inductors and coupling

21. A $5\,\mu$F capacitor is at 12 V. Find its charge.

22. Capacitors 6 and $3\,\mu$F in series: equivalent capacitance?
23. Capacitors 10 and $20\,\mu$F in parallel: equivalent capacitance?
24. A $20\,\mu$F capacitor is charged to 50 V. Find energy.
25. A $5\,\mu$F capacitor voltage rises 3 V per millisecond. Find current.
26. A 4 H inductor carries 3 A. Find energy.
27. A coupled pair yields 10 H aiding and 6 H opposing. Find mutual inductance.
28. For $L_1 = 9$ H, $L_2 = 4$ H and $k = 0.5$, find M.
29. A 200-turn coil links 0.004 Wb per turn at 2 A. Find self-inductance (linear model).
30. Flux through a 1200-turn coil changes uniformly by 0.005 Wb in 0.01 s. Find EMF magnitude.

C. Magnetic circuits and induction

31. Magnetic flux is 0.002 Wb and reluctance is 5×10^5 At/Wb. Find MMF.
32. An area 10^{-3} m^2 is normal to uniform 0.4 T flux density. Find flux.
33. A 300-turn solenoid carries 2 A and has 0.4 m mean magnetic path. Find H.
34. Find ideal inductance: $N = 300$, $\mu_r = 1000$, $A = 2 \times 10^{-4}$ m^2, $\ell = 0.3$ m.
35. A linear core has $B = 0.6$ T for $H = 300$ A/m. Find absolute permeability.
36. For a 400 turn coil with 0.8 A excitation, find total MMF.
37. A core has $B = 1.2$ T over 2 cm^2. Find flux in mWb.
38. A 100 turn coil has a flux-linkage change of 0.2 Wb-turn over 0.04 s. Find average induced voltage.

D. AC waveforms, phasors and power

39. A sine wave has 325.27 V peak. Find RMS.
40. Frequency 100 Hz: find angular frequency.
41. A pure sine current has a peak of 20 A. Find RMS.
42. An RL series circuit has $R = 6\,\Omega$, $X_L = 8\,\Omega$. Find magnitude and pf.
43. A series RC circuit has $R = 8\,\Omega$, $X_C = 6\,\Omega$. Find pf.
44. A series RLC circuit has $R = 5\,\Omega$, $X_L = 15\,\Omega$, $X_C = 9\,\Omega$. Find $|Z|$.
45. Parallel $R = 10\,\Omega$ and ideal capacitor $X_C = 10\,\Omega$ share 100 V RMS. Find total current magnitude.
46. Apparent power 200 VA and lagging pf 0.8: find P and Q.
47. At 60 Hz a $100\,\mu$F capacitor is connected alone. Find X_C.
48. A 100 V RMS AC source is across $R = 5\,\Omega$ and $X_L = 5\,\Omega$ in series. Find mean power.

E. Resonance and first-order transients

49. What capacitance resonates with $L = 0.1$ H at 50 Hz in series?

50. At series resonance, $V = 100$ V, $R = 4\,\Omega$, $X_L = 20\,\Omega$. Find capacitor voltage.

51. Series RLC circuit has $R = 10\,\Omega$, $L = 0.1$ H. Compute approximate bandwidth in Hz.

52. Load real power 500 W at 0.8 lagging pf. Find capacitive vars required for unity pf.

53. What is the time constant of $R = 10\,\text{k}\Omega$ with $C = 10\,\mu$F?

54. A zero-voltage capacitor charges toward 10 V. Find its voltage at $t = \tau$.

55. An inductor starts with 5 A and decays freely. Find current after one time constant.

56. At series RLC resonance, what are impedance, power factor, and source reactive power?

F. Balanced three-phase circuits

57. A 400 V *line* supply feeds a balanced star load of $20\,\Omega$ per phase, purely resistive. Find line current.

58. Three $20\,\Omega$ resistors form a delta across 400 V *line*. Find line current.

59. A balanced 400 V line, 10 A line, pf 0.8 load: find 3-phase real power.

60. Three $30\,\Omega$ arms connected as a balanced Y to 300 V line: find power.

61. The same three $30\,\Omega$ arms form delta on the same 300 V line: power?

62. In balanced 3-phase two-wattmeter method, $W_1 = 6$ kW and $W_2 = 2$ kW, larger one lagging convention. Find pf.

63. A 400 V line, 20 A line balanced load draws pf 0.5. Find three-phase real power.

64. A 3-phase Δ capacitor bank must provide 3000 var on 400 V line at 50 Hz. Find capacitance per leg.

17.2 Answers and the method in one line

Check the *equation* as well as the final answer. Circle any item where your procedure differed, even if the number happened to match.

A. DC laws, resistive networks and theorems

1. $I = 12/3 = \boxed{4\,\text{A}}$. Use Ohm's law.

2. $I = 48/(4 + 12) = \boxed{3\,\text{A}}$. Series resistances add.

3. $R = 20(30)/(20 + 30) = \boxed{12\,\Omega}$.

4. $V_4 = 18(4/6) = \boxed{12\,\text{V}}$.

5. $I_4 = 5(6/10) = \boxed{3\,\text{A}}$.

6. $A' = A/2$; $R'/R = 2/(1/2) = \boxed{4}$.

7. $R_{70} = 10[1 + 0.004(50)] = \boxed{12\,\Omega}$.

8. $I = \sqrt{P/R} = \sqrt{4} = \boxed{2\,\text{A}}$.
9. $I = 2 - 1.2 - 0.3 = \boxed{0.5\,\text{A}}$ by KCL.
10. $I = (10 - 0)/5 = \boxed{2\,\text{A}}$.
11. $I = 9/(3 + 6) = \boxed{1\,\text{A}}$.
12. $P_{\text{max}} = 24^2/(4 \cdot 6) = \boxed{24\,\text{W}}$.
13. $V_{oc} = I_N R_N = \boxed{20\,\text{V}}$.
14. $I_N = 10/2 = \boxed{5\,\text{A}}$ in parallel with $2\,\Omega$.
15. $4/8 = 6/R_4 \Rightarrow \boxed{R_4 = 12\,\Omega}$.
16. $R_Y = R_\Delta/3 = \boxed{7\,\Omega}$ per branch.
17. $R_\Delta = 3R_Y = \boxed{15\,\Omega}$ per branch.
18. $Z = 1/Y = (4 - \text{j}3)/25$; $\boxed{R = 0.16\,\Omega}$.
19. $P = V^2/R = 40000/100 = \boxed{400\,\text{W}}$.
20. $R = 15(30)/45 = \boxed{10\,\Omega}$.

B. Capacitors, inductors and coupling

21. $Q = CV = 5 \cdot 12 = \boxed{60\,\mu\text{C}}$.
22. $C = 6(3)/(6 + 3) = \boxed{2\,\mu\text{F}}$.
23. $C = 10 + 20 = \boxed{30\,\mu\text{F}}$.
24. $W = \frac{1}{2}CV^2 = \boxed{0.025\,\text{J}}$.
25. $i = C\,dv/dt = 5 \times 10^{-6}(3000) = \boxed{15\,\text{mA}}$.
26. $W = \frac{1}{2}Li^2 = \frac{1}{2}(4)(9) = \boxed{18\,\text{J}}$.
27. $M = (10 - 6)/4 = \boxed{1\,\text{H}}$.
28. $M = k\sqrt{L_1 L_2} = 0.5(6) = \boxed{3\,\text{H}}$.
29. $L = N\Phi/I = 200(0.004)/2 = \boxed{0.4\,\text{H}}$.
30. $|e| = N|\Delta\Phi/\Delta t| = 1200(.005/.01) = \boxed{600\,\text{V}}$.

C. Magnetic circuits and induction

31. $\mathcal{F} = \Phi\mathcal{R} = .002(500000) = \boxed{1000\,\text{At}}$.
32. $\Phi = BA = 0.4(10^{-3}) = \boxed{4 \times 10^{-4}\,\text{Wb}}$.
33. $H = NI/\ell = 600/.4 = \boxed{1500\,\text{A/m}}$.
34. $L = \mu_0\mu_r N^2 A/\ell \approx \boxed{0.0754\,\text{H}}$.

35. $\mu = B/H = \boxed{0.002\,\text{H/m}}$.

36. $NI = 400(0.8) = \boxed{320\,\text{At}}$.

37. $\Phi = 1.2(2 \times 10^{-4}) = \boxed{0.24\,\text{mWb}}$.

38. $|e| = |\Delta\lambda|/\Delta t = .2/.04 = \boxed{5\,\text{V}}$.

D. AC waveforms, phasors and power

39. $V_{rms} = V_m/\sqrt{2} \approx \boxed{230\,\text{V}}$.

40. $\omega = 2\pi f = \boxed{200\pi\,\text{rad/s}}$.

41. $I_{rms} = 20/\sqrt{2} \approx \boxed{14.14\,\text{A}}$.

42. $|Z| = \sqrt{36 + 64} = \boxed{10\,\Omega}$; pf $= 6/10 = \boxed{0.6}$ lag.

43. $|Z| = 10\,\Omega$; $\boxed{\text{pf} = 0.8 \text{ leading}}$.

44. $|Z| = \sqrt{5^2 + (15-9)^2} = \boxed{\sqrt{61} \approx 7.81\,\Omega}$.

45. $\underline{I} = 10 + \text{j}10$ A; $\boxed{|I| = 14.14\,\text{A}}$, pf 0.707 leading.

46. $P = 200(.8) = \boxed{160\,\text{W}}$; $Q = 200(.6) = \boxed{120\,\text{var}}$.

47. $X_C = 1/(2\pi 60 \cdot 10^{-4}) \approx \boxed{26.53\,\Omega}$.

48. $P = V^2R/(R^2 + X_L^2) = 10000(5)/50 = \boxed{1000\,\text{W}}$.

E. Resonance and first-order transients

49. $C = 1/[(2\pi 50)^2(0.1)] \approx \boxed{101.3\,\mu\text{F}}$.

50. $I = V/R = 25$ A; $V_C = IX_L = \boxed{500\,\text{V}}$.

51. $BW = R/(2\pi L) = 10/(.2\pi) = \boxed{15.92\,\text{Hz}}$.

52. $Q_c = P\tan(\cos^{-1} 0.8) = 500(.75) = \boxed{375\,\text{var}}$.

53. $\tau = RC = 10000(10^{-5}) = \boxed{0.1\,\text{s}}$.

54. $v = 10(1 - e^{-1}) \approx \boxed{6.32\,\text{V}}$.

55. $i = 5e^{-1} \approx \boxed{1.84\,\text{A}}$.

56. $Z = \boxed{R}$, pf $= \boxed{1}$, net $Q = \boxed{0}$.

F. Balanced three-phase circuits

57. $I_L = 400/(\sqrt{3} \cdot 20) = \boxed{11.55\,\text{A}}$.

58. $I_{ph} = 400/20 = 20$ A; $I_L = \sqrt{3}(20) = \boxed{34.64\,\text{A}}$.

59. $P = \sqrt{3}(400)(10)(0.8) \approx \boxed{5.54\,\text{kW}}$.

60. $P_Y = V_L^2/R = 300^2/30 = \boxed{3\,\text{kW}}$.

61. $P_\Delta = 3V_L^2/R = 3(300^2)/30 = \boxed{9\,\text{kW}}$.

62. $\tan\phi = \sqrt{3}(6-2)/(6+2) = \sqrt{3}/2$; $\boxed{\text{pf} \approx 0.756}$ lag.

63. $P = \sqrt{3}(400)(20)(.5) \approx \boxed{6.93\,\text{kW}}$.

64. $C_\Delta = 3000/[3(2\pi 50)(400)^2] \approx \boxed{19.89\,\mu\text{F}}$.

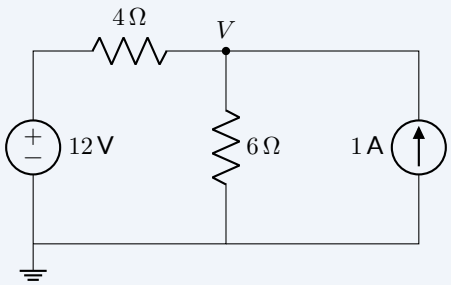
UNIT 18

Twelve multi-step problems with full methods

These original integrative examples are a level above quick formula substitutions. Each solution makes the model-selection decision explicit.

Worked example **Challenge 1 — nodal analysis with a source and injected current**

A node V is connected through $4\,\Omega$ to a known 12 V node, through $6\,\Omega$ to ground, and receives 1 A from an independent current source directed upward from ground. Find V and the current in each resistor.



Step 1 — select nodal analysis. Only one unknown node voltage exists; a current source enters it.
Step 2 — KCL: currents leaving through resistors equal 1 A entering:

$$\frac{V-12}{4}+\frac{V}{6}=1.$$

Step 3 — solve: multiply by 12 to obtain $3V-36+2V=12$, hence $5V=48$ and $V=9.6$ V.
Step 4 — verify: current from 12 V node to V is $(12-9.6)/4=0.6$ A, through $6\,\Omega$ is $9.6/6=1.6$ A; at node $0.6+1=1.6$ A.
Answer: $\boxed{V=9.6\,\text{V},\quad I_{4\Omega}=0.6\,\text{A},\quad I_{6\Omega}=1.6\,\text{A}}$.

Worked example **Challenge 2 — two-mesh simultaneous equations**

Clockwise mesh currents i_1, i_2 pass through $4\,\Omega$ in mesh 1, $3\,\Omega$ in mesh 2, and a shared $2\,\Omega$ resistor. A 10 V source drives mesh 1; a 5 V source drives mesh 2 with the chosen clockwise reference. Solve both mesh currents.

$$\underbrace{(4+2)i_1-2i_2=10}_{\text{mesh 1}},\qquad \underbrace{-2i_1+(3+2)i_2=5}_{\text{mesh 2}}.$$

From the first, $3i_1-i_2=5$, so $i_2=3i_1-5$. Substitute in the second:

$$-2i_1+5(3i_1-5)=5 \implies 13i_1=30.$$

Thus $i_1=30/13=2.308$ A and $i_2=25/13=1.923$ A. Shared branch current is their *difference* $i_1-i_2=5/13$ A.
Answer: $\boxed{i_1=30/13\,\text{A},\quad i_2=25/13\,\text{A},\quad i_{shared}=5/13\,\text{A}}$.

Worked example **Challenge 3 — Thevenin from open circuit and a load**

An accessible black-box DC network measures $V_{oc} = 24$ V. When a $6\,\Omega$ load is attached, the terminal voltage drops to 12 V. Find R_{th} and the load for maximum power. **Step 1:** load current $I = 12/6 = 2$ A.
Step 2: internal drop $24 - 12 = 12$ V, so $R_{th} = 12/2 = 6\,\Omega$.
Step 3: $R_{L,opt} = R_{th} = 6\,\Omega$ and $P_{max} = 24^2/(4 \times 6) = 24$ W.
Answer: $\boxed{R_{th} = 6\,\Omega, \quad R_{L,opt} = 6\,\Omega, \quad P_{max} = 24\,\text{W}}$.

Worked example **Challenge 4 — inserting dielectric in two separate conditions**

An initially $10\,\mu$F capacitor at 100 V is completely filled with dielectric constant $k = 3$. Compare results when left attached to a 100 V ideal source and when isolated before insertion. **Common:** $C' = 30\,\mu$F; initial $Q = CV = 1000\,\mu$C; initial $W = \frac{1}{2}(10 \times 10^{-6})(100)^2 = 0.05$ J.
Connected: $V' = 100$ V; $Q' = 30\,\mu\text{F} \cdot 100\,\text{V} = 3000\,\mu$C; $W' = 0.15$ J.
Isolated: $Q' = 1000\,\mu$C, $V' = Q'/C' = 100/3 \approx 33.3$ V; $W' = 0.05/3 \approx 0.0167$ J.
Answer: $\boxed{W_{connected} = 0.15\,\text{J}, \quad W_{isolated} \approx 0.0167\,\text{J}}$.

Worked example **Challenge 5 — coupling coefficient to aiding/opposing values**

Two inductors have $L_1 = 12$ mH and $L_2 = 20$ mH, with coupling coefficient $k = 0.5$. Compute M, L_+, L_-.

$$M = k\sqrt{L_1 L_2} = 0.5\sqrt{12(20)} \approx 7.746\,\text{mH}.$$

$$L_+ = 12 + 20 + 2(7.746) = 47.492\,\text{mH},$$

$$L_- = 12 + 20 - 2(7.746) = 16.508\,\text{mH}.$$

Sanity: $L_+ - L_- = 4M$, and both equivalent values are positive.
Answer: $\boxed{M \approx 7.746\,\text{mH}, \quad L_+ \approx 47.49\,\text{mH}, \quad L_- \approx 16.51\,\text{mH}}$.

Worked example **Challenge 6 — reluctance with one air gap**

A magnetic core has mean iron length 0.5 m, uniform area $4\,\text{cm}^2$, relative permeability 1000, and one air gap of 1 mm. A 500-turn winding carries 2 A. Neglect leakage and fringing. Find flux and gap flux density.

$$A = 4 \times 10^{-4}\,\text{m}^2, \quad \mathcal{F} = NI = 1000\,\text{At},$$

$$\mathcal{R}_i = \frac{0.5}{(4\pi \times 10^{-7})(1000)(4 \times 10^{-4})} \approx 9.95 \times 10^5\,\text{At/Wb},$$

$$\mathcal{R}_g = \frac{0.001}{(4\pi \times 10^{-7})(4 \times 10^{-4})} \approx 1.989 \times 10^6\,\text{At/Wb}.$$

$$\Phi = \frac{1000}{\mathcal{R}_i + \mathcal{R}_g} \approx 3.35 \times 10^{-4}\,\text{Wb}, \quad B_g = \Phi/A \approx 0.838\,\text{T}.$$

Pitfall: the tiny air gap contributes about *twice* the reluctance of the long high-permeability iron section. Do not discard gap reluctance.
Answer: $\boxed{\Phi \approx 0.335\,\text{mWb}, \quad B_g \approx 0.838\,\text{T}}$.

Worked example **Challenge 7 — complex impedance to current, P, Q and pf**

A single-phase load at 230 V RMS is $Z = 8 + \text{j}6\,\Omega$. Find rectangular current, RMS magnitude, real power, reactive power, and pf.

$$\underline{I} = \frac{230}{8+\text{j}6} = \frac{230(8-\text{j}6)}{100} = 18.4 - \text{j}13.8\,\text{A}.$$

$$|I| = 23\,\text{A}, \quad \cos\phi = \frac{8}{10} = 0.8 \text{ lagging}.$$

$$P = I^2R = 23^2(8) = 4232\,\text{W}, \qquad Q = I^2X = 23^2(6) = 3174\,\text{var}.$$

Answer: $\boxed{|I| = 23\,\text{A},\ P = 4.232\,\text{kW},\ Q = 3.174\,\text{kvar},\ \text{pf} = 0.8 \text{ lagging}}$.

Worked example **Challenge 8 — power factor correction to a specified target**

A single-phase load uses 10 kW at 0.6 pf lagging. How many capacitive kvar improve pf to 0.9 lagging at unchanged real power?

$$\tan\phi_1 = \frac{\sqrt{1-0.6^2}}{0.6} = \frac{4}{3} = 1.3333,$$

$$\tan\phi_2 = \frac{\sqrt{1-0.9^2}}{0.9} \approx 0.4843.$$

$$Q_c = 10(1.3333 - 0.4843) \approx 8.49\,\text{kvar}.$$

Variant: if the question additionally asks for capacitance on a 230 V, 50 Hz supply, $C = Q_c/(\omega V^2)$.
Answer: $\boxed{Q_c \approx 8.49\,\text{kvar}}$.

Worked example **Challenge 9 — periodic waveform with DC and harmonics**

Compute RMS of $v(t) = 5 + 3\sin\omega t + 4\sin 2\omega t$ volts over a complete common period.

$$V_{rms}^2 = 5^2 + \frac{3^2}{2} + \frac{4^2}{2} = 25 + 4.5 + 8 = 37.5.$$

$$V_{rms} = \sqrt{37.5} \approx 6.124\,\text{V}.$$

The DC level contributes its *square*; do not treat it as a sinusoidal peak.
Answer: $\boxed{V_{rms} \approx 6.124\,\text{V}}$.

Worked example **Challenge 10 — a complex delta-connected three-phase load**

Three identical $Z_\Delta = 2 + \text{j}4\,\Omega$ branches connect in delta to a balanced 415 V line supply. Find line current, pf, and three-phase real power.

$$|Z| = \sqrt{2^2+4^2} = \sqrt{20}\,\Omega, \qquad I_{ph} = \frac{415}{\sqrt{20}} \approx 92.80\,\text{A}.$$

$$I_L = \sqrt{3}I_{ph} \approx 160.7\,\text{A}, \qquad \cos\phi = \frac{2}{\sqrt{20}} \approx 0.4472 \text{ lagging}.$$

$$P = 3I_{ph}^2R = 3\left(\frac{415^2}{20}\right)2 = 51667.5\,\text{W}.$$

Answer: $\boxed{I_L \approx 160.7\,\text{A}, \quad P \approx 51.67\,\text{kW}}$.

Worked example **Challenge 11 — transient terminal value and monotonicity**

An initially uncharged $20\,\mu$F capacitor is connected to a 20 V source through $5\,\text{k}\Omega$ at $t = 0$. Determine (i) the time for the voltage to reach 90% of final, (ii) source current at that time.

$$\tau = RC = 5000(20 \times 10^{-6}) = 0.1\,\text{s}.$$

$$0.9 = 1 - e^{-t/\tau} \Rightarrow e^{-t/\tau} = 0.1 \Rightarrow \boxed{t = \tau \ln 10 \approx 0.2303\,\text{s}}.$$

$$i(t) = \frac{V_s}{R} e^{-t/\tau} = \frac{20}{5000}(0.1) = 0.0004\,\text{A}.$$

Answer: $\boxed{t \approx 0.2303\,\text{s}, \quad i = 0.4\,\text{mA}}$.

Worked example **Challenge 12 — three-phase capacitor bank comparison**

A balanced load needs 3 kvar of capacitive compensation from a bank supplied at 400 V line, 50 Hz. Find capacitance *per phase* for equivalent star and delta banks.

$$\omega = 2\pi(50) \approx 314.159\,\text{rad/s}, \quad C_\Delta = \frac{3000}{3\omega(400)^2} \approx 19.89\,\mu\text{F}.$$

$$C_Y = \frac{3000}{\omega(400)^2} \approx 59.68\,\mu\text{F}.$$

Reason: star capacitor phase voltage is $V_L/\sqrt{3}$, while delta phase voltage is V_L; each bank must deliver the *same total* vars.

Answer: $\boxed{C_\Delta \approx 19.89\,\mu\text{F}, \quad C_Y \approx 59.68\,\mu\text{F}}$.

UNIT 19

The complete last-day formula compendium

This is the formula section to revisit before every sectional mock. Understand the preceding derivations first; memorizing this page alone will not solve a differently drawn circuit.

19.1 Fundamental quantities and resistor networks

$$I = \frac{dq}{dt}, \qquad V = \frac{W}{q},$$

$$P = VI = I^2R = \frac{V^2}{R}, \qquad E = Pt,$$

$$R = \frac{\rho \ell}{A}, \qquad G = \frac{1}{R},$$

$$\sigma = \frac{1}{\rho}, \qquad R_T = R_0[1 + \alpha_0(T - T_0)],$$

$$R_{series} = \sum_i R_i, \qquad R_{parallel}^{-1} = \sum_i \frac{1}{R_i},$$

$$R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}, \qquad V_k = V \frac{R_k}{\sum R_i},$$

$$I_1 = I \frac{R_2}{R_1 + R_2}, \qquad \sum I_{node} = 0, \qquad \sum V_{loop} = 0.$$

Key result

Delta-to-star, general case: let delta sides be R_{ab}, R_{bc}, R_{ca} and $S = R_{ab} + R_{bc} + R_{ca}$.

$$R_a = \frac{R_{ab}R_{ca}}{S}, \quad R_b = \frac{R_{ab}R_{bc}}{S}, \quad R_c = \frac{R_{bc}R_{ca}}{S}.$$

Star-to-delta: $R_{ab} = (R_aR_b + R_bR_c + R_cR_a)/R_c$ (opposite star resistor in denominator), cyclically.

19.2 Network analysis and theorems

$$V_{th} = V_{oc}, \qquad I_N = I_{sc}, \qquad R_{th} = R_N = V_{oc}/I_{sc} \quad \text{(where valid)},$$

$$I_L = \frac{V_{th}}{R_{th} + R_L}, \qquad V_L = V_{th} \frac{R_L}{R_{th} + R_L},$$

$$R_{L,opt} = R_{th}, \qquad P_{max} = \frac{V_{th}^2}{4R_{th}}, \qquad \eta_{opt} = 50\% \text{ for resistive Thevenin model.}$$

Independent ideal voltage source deactivated $\Rightarrow$ short; independent ideal current source deactivated $\Rightarrow$ open; *dependent sources stay in the circuit*. For a dependent-source resistance, apply a test source and use $R_{th} = V_{test}/I_{test}$.

19.3 Capacitors, dielectrics and transient response

$$Q = CV, \qquad i_C = C\,dv_C/dt, \qquad C_{plates} = \varepsilon A/d,$$
$$C_{parallel} = \sum C_i, \qquad C_{series}^{-1} = \sum C_i^{-1}, \qquad W_C = \tfrac{1}{2}CV^2,$$
$$\tau_{RC} = R_{th}C, \qquad v_C(t) = v_C(\infty) + [v_C(0^+) - v_C(\infty)]e^{-t/\tau_{RC}},$$
$$\text{at } t = \tau: \quad 1 - e^{-1} = 0.632, \qquad \text{at } t = 3\tau: \quad 1 - e^{-3} \approx 0.950.$$

19.4 Inductors, magnetism and mutual coupling

$$\lambda = N\Phi = Li, \qquad v_L = L\,di_L/dt, \qquad W_L = \tfrac{1}{2}LI^2,$$
$$\mathcal{F} = NI = \Phi\mathcal{R}, \qquad \mathcal{R} = \ell/(\mu A), \qquad \Phi = BA, \qquad B = \mu H,$$
$$L = \mu N^2 A/\ell, \qquad M = k\sqrt{L_1 L_2}, \qquad |k| \le 1,$$
$$L_{aid} = L_1 + L_2 + 2M, \qquad L_{opp} = L_1 + L_2 - 2M, \qquad M = (L_{aid} - L_{opp})/4,$$
$$\tau_{RL} = L/R_{th}, \qquad i_L(t) = i_L(\infty) + [i_L(0^+) - i_L(\infty)]e^{-t/\tau_{RL}}.$$

19.5 AC waveforms, impedance and power

$$\omega = 2\pi f, \qquad T = 1/f, \qquad X_L = \omega L, \qquad X_C = 1/(\omega C),$$
$$V_{rms,sine} = V_m/\sqrt{2}, \qquad V_{avg,half-sine} = 2V_m/\pi, \qquad k_f = 1.11, \qquad k_p = 1.414,$$
$$Z_R = R, \qquad Z_L = \mathrm{j}X_L, \qquad Z_C = -\mathrm{j}X_C, \qquad Z_{series} = \sum Z_i,$$
$$Y_{parallel} = \sum_i 1/Z_i, \qquad |R + \mathrm{j}X| = \sqrt{R^2 + X^2}, \qquad \phi = \tan^{-1}(X/R),$$
$$\underline{S} = \underline{V}\underline{I}^* = P + \mathrm{j}Q, \qquad P = VI\cos\phi, \qquad Q = VI\sin\phi, \qquad |S| = VI,$$
$$\text{pf} = P/|S|, \qquad P_R = I^2R, \qquad Q_L = +I^2X_L, \qquad Q_C = -I^2X_C.$$

19.6 Resonance, quality factor and correction

$$\omega_0 = 1/\sqrt{LC}, \qquad f_0 = 1/(2\pi\sqrt{LC}),$$
$$\text{series RLC: } Z_0 = R, \qquad I_0 = V/R, \qquad Q_s = \omega_0 L/R = 1/(\omega_0 CR),$$
$$BW_{rad/s} = R/L, \qquad BW_{Hz} = R/(2\pi L), \qquad Q_s = f_0/BW_{Hz},$$
$$V_L\big|_0 = QV, \qquad V_C\big|_0 = QV, \qquad Q_{c,needed} = P(\tan\phi_1 - \tan\phi_2),$$
$$C_{1\phi} = Q_c/(\omega V^2).$$

19.7 Balanced three-phase formulas

$$\text{Y: } V_L = \sqrt{3}V_{ph}, \qquad I_L = I_{ph},$$
$$\Delta\text{: } V_L = V_{ph}, \qquad I_L = \sqrt{3}I_{ph},$$
$$P_{3\phi} = \sqrt{3}V_L I_L \cos\phi, \qquad Q_{3\phi} = \sqrt{3}V_L I_L \sin\phi,$$
$$S_{3\phi} = \sqrt{3}V_L I_L, \qquad Z_Y = Z_\Delta/3 \text{ (balanced equivalent)},$$
$$P_{\Delta,\text{ same } Z}/P_{Y,\text{ same } Z} = 3, \qquad C_\Delta = Q_c/(3\omega V_L^2),$$
$$C_Y = Q_c/(\omega V_L^2), \qquad \tan\phi = \sqrt{3}(W_1 - W_2)/(W_1 + W_2).$$

SSC JE trap / important distinction

Assumptions matter. The $\sqrt{3}$ line relations above require balanced systems. The two-wattmeter pf equation requires a balanced sinusoidal load and consistent wattmeter sign conventions. In resonant parallel networks with coil resistance, the frequency of *maximum input impedance* may differ from the simple lossless $1/(2\pi\sqrt{LC})$ approximation.

19.8 Units and constants you must convert instantly

Prefix	Multiplier	Conversion	Value
kilo (k)	10^3	1 kWh	3.6×10^6 J
mega (M)	10^6	1 μF	10^{-6} F
milli (m)	10^{-3}	1 mH	10^{-3} H
micro (μ)	10^{-6}	1 cm^2	10^{-4} m^2
nano (n)	10^{-9}	1 mm	10^{-3} m
pico (p)	10^{-12}	ε_0	8.854×10^{-12} F/m

Key result

Exam memory board: $\mu_0 = 4\pi \times 10^{-7}$ H/m; $\sqrt{2} \approx 1.414$; $\sqrt{3} \approx 1.732$; $\pi \approx 3.1416$; $e^{-1} \approx 0.3679$; $\tan(\cos^{-1} 0.8) = 0.75$; $\tan(\cos^{-1} 0.6) = 4/3$.

UNIT 20

Revision sequence, practice targets and error log

20.1 A repeatable 10-day chapter plan

Day	Study and derive	Test yourself
1	Units, resistance, sources, wire problems	8–12 basic DC problems
2	KCL/KVL, nodal method, mesh method	10 networks, include a supernode and supermesh
3	Thevenin, Norton, superposition, maximum power	12 mixed source/network models
4	Capacitance, dielectrics, stored energy	10 charge, voltage and energy variations
5	Inductors, mutual coupling, magnetic circuits	12 coils, gap, MMF and reluctance variations
6	Waveform RMS/average, phasors, RLC	12 waveform and impedance questions
7	AC real/reactive/complex power, PF correction	10 leading/lagging and kvar questions
8	Series/parallel resonance, RL/RC transients	12 resonance and time-constant questions
9	Star, delta, three-phase power and two-wattmeter	12 three-phase questions
10	All identified PYQs and 64 original drills	Timed sectional mock and comprehensive error analysis

20.2 How to review a wrong attempt

For each incorrect question, write (a) what clue should have identified the method; (b) the precise mistaken step; (c) the corrected equation; and (d) a *new* variant you can solve without copying. Reattempt missed questions after 1, 3 and 7 days; the purpose is to remove a repeatable error, not merely remember the previous number.

20.3 Common traps, compressed

1. **Capacitor and inductor at $t = 0^+$.** Capacitor voltage and inductor current normally stay continuous; the DC long-time open/short result is *not* the immediate switched behavior.
2. **Power under superposition.** Sum source-produced currents or voltages first and square the final combined value; do not add powers from individual-source circuits.
3. **Complex sign.** $X > 0$ inductive/lags; $X < 0$ capacitive/leads, using standard passive load convention.
4. **Series versus parallel.** Add series impedances Z; add parallel admittances Y; never interchange $\Re(Y)$ with the real part of Z.

5. **Magnetic gap.** Air-gap MMF can dominate even when the gap is much shorter than the iron length.
6. **Mutual coil sign.** Aiding $+2M$ and opposing $-2M$; check dot convention before writing KVL for coupled windings.
7. **Real/complex power.** P is W, Q is var and $|S|$ is VA; Q is negative for a capacitive load under the usual convention.
8. **Line/phase confusion.** Y and Δ have different line-to-phase relationships; identify the connection before calculating current or per-phase power.
9. **RMS/peak confusion.** Heating and common power formulas require RMS values unless otherwise stated; a sinusoidal peak is $\sqrt{2}$ times RMS.
10. **Too-fast formula selection.** Draw node voltages and directions. A correct equation written with a provisional current direction always gives a consistent sign.

UNIT 21

Sources and independent-verification notes

SSC official syllabus (2026). *Notice of Junior Engineer Examination 2026*, section 14.3.2 (printed page 26), Fundamentals, Network Solution Methods, Network Theorems, Magnetic Circuits and AC Fundamentals.

Open the SSC JE 2026 official notification (PDF).

21.1 Sources consulted for PYQ identifiers

1. **2024 June 5, Shift 3, Paper I Electrical.** Source: Prepp reproduction of the full question paper with response material. Refer to items Q14 (capacitor derivative), Q15 (resonant capacitor voltage), Q78 (harmonic RMS). Open paper PDF.
2. **2024 June 6, Shift 2, Paper I Electrical.** Q40 (superposition), Q83 (KVL). Open paper PDF.
3. **2024 June 7, Shift 3, Paper I Electrical.** Q44 (mutual inductance), Q86 (fringing), Q87 (coupling bound). Open paper PDF.
4. **2024 November 6, Paper II Electrical.** Q16 (Thevenin load current). Open paper PDF.
5. **2023 October 11, Shift 3, Electrical.** Q20 (RL impedance), Q23 (energy), Q80 (sine RMS), Q81 (coupling), Q82 (LED series resistance), Q83 (plate capacitance). Open paper PDF.
6. **2023 October 9, Shift 3, Electrical.** Q5 (apparent power), Q28 (series resistance from admittance). Open paper PDF.
7. **2025 December 4, Shift 2, Paper I Electrical.** Public searchable transcription identifying Q101–106, Q141–143, Q149, Q151–153, Q171–172, Q187–189, Q192–195. This is a third-party transcription, not an SSC-hosted paper; independently check any disputed figures using SSC candidate-response access. Open public transcription. A second paper index also lists that date and shift.

21.2 Worked PYQ editorial standard

The question descriptions in this book are short paraphrases of the corresponding source items, not reproductions of all original answer options. Displayed solutions use standard electrical engineering equations, derived in earlier chapters. Any discrepancy between a candidate's *selected* answer and the actual physical result has been checked independently (notably the 2024 June 5 resonance question). The practice bank is newly constructed and labeled separately. The official SSC notification determines *2026 syllabus coverage*; old PYQs supply examples, not a promise of future topic frequency.

End of master handbook

Keep revising the method, not just the final number.